KOZMA'S CENTRALLY EXCITED WALK CONVERGES TO A EUCLIDEAN BALL

AHMED BOU-RABEE, TUAN-MINH NGUYEN, YUVAL PERES, AND ARVIND SINGH

Abstract. We study nearest-neighbor walks whose first departure from each site has an inward mean prescribed by a convex gauge, and whose later departures are simple random walk. The first-departure laws may depend on the past, and the drift strength may have spatial defects of codimension one. One potential argument gives an almost-sure filled limit shape, uniform conical local times, and recurrence. In the Euclidean case it retains the sharp planar fluctuation exponent, polylogarithmic radius bounds in higher dimensions, and the local-time limit laws. A quadratic martingale gives Gaussian and iterated-logarithm laws for a weighted radial moment for every convex body. An energy identity additionally controls the number of sites with small local time. The proof combines exit-time stopping, contact-point positivity, and a finite-scale calibration, and includes the three previously studied models. A squared-Green identity makes the variance estimates local, and Green differences give bulk oscillation lower bounds for every convex gauge. A sharp anisotropic cone inequality also strengthens the Sobolev estimate used in the calibration approach.

1. Introduction

An inward-excited walk is attracted toward a prescribed center only when it leaves a site for the first time. After that departure the site has no remaining bias. In the Euclidean example, the first step from $x \neq 0$ has mean $-\kappa x/|x|$, while every later step is uniform among the $2d$ neighbors. The path therefore remembers where it has been, even though its motion at a previously departed site is simple random walk. Figure 1 illustrates this rule.

Kozma asked whether this attraction produces a ball of radius of order $n^{1/(d+1)}$ after n steps [30, 31]. The three papers considered here establish shape theorems for related inward drifts. The present aim is to give one proof that explains both the common behavior and the different limiting shapes, while retaining the fluctuation and sharpness results.

The local times provide the useful description of the growth. A site deep inside the range has been visited many times, whereas a site near the boundary has only recently supplied its inward step. The limiting profile is a cone: its height is proportional to the distance from the boundary measured in the appropriate gauge. Integrating a cone of height r over a body of diameter r gives a mass of order r^{d+1}. Since the sum of the local times is exactly n, this identifies the scale $r \asymp n^{1/(d+1)}$. The potential identity below determines both the cone and its multiplicative constant.

The range of an inward-excited walk is determined at first order by its mean excitation. The probabilities realizing that mean, and spatial changes confined to sufficiently thin sets, affect only lower-order terms. We make this principle quantitative. The argument unifies Singh [42], Bou-Rabee and Peres [9], and Nguyen [37]; all three models are special cases of the same assumptions below.

1.1. The model and its geometry.

A convex gauge records the shape toward which the drift points. Its subgradient at x is a normal to a supporting hyperplane of the level set through x; it need not be parallel to x. For a Euclidean ball it is $x/|x|$, for an ℓ^1 ball one may take the coordinate signs, and for a body with corners there can be several choices. The argument will be uniform over those choices. This is why the model is stated using subgradients rather than a smooth boundary.

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Fix $d \geq 2$. Let $K \subset \mathbb{R}^d$ be compact and convex, with 0 in its interior, and let $\rho(x) = \inf\{t > 0 : x \in tK\}$ be its gauge. Write $V = |K|$ and $\omega_d = |B(0,1)|$. Symmetry and smoothness of K are not required. Fix $\kappa > 0$, subgradients $\xi(x) \in \partial\rho(x)$ for $x \in \mathbb{Z}^d \setminus \{0\}$, and weights $\lambda(x) \geq \lambda_- > 0$. Assume that, for a constant $A < \infty$,

$$\sum_{x \in \mathbb{Z}^d \cap B(y,s)} |\lambda(x) - 1| \leq A s^{d-1} \qquad (y \in \mathbb{R}^d,\ s \geq 1). \tag{1.1}$$

Set $\xi(0) = 0$ and $\lambda(0) = 1$. Taking $s = 1$ above gives $\lambda(x) \leq 1 + A$, so no separate upper bound is needed.

The walk starts at $X_0 = 0$. On its first departure from x, its conditional step distribution, given the entire past, can be any distribution on $\{\pm e_1, \ldots, \pm e_d\}$ with mean

$$-\kappa\lambda(x)\xi(x). \tag{1.2}$$

On every later departure it chooses its neighbor uniformly. The first departure from zero may have any mean-zero nearest-neighbor law. Only existence of the distributions in (1.2) is assumed; in particular no lower bound on individual probabilities is required. The fields λ and ξ are deterministic. Allowing the realizing distributions to depend on the past is useful because every estimate below uses their conditional means and bounded steps.

Condition (1.1) allows the drift strength to differ from its nominal value on a thin set. The sum in any ball grows like its surface area, rather than its volume. Coordinate hyperplanes are an important example: they are precisely where Nguyen's transition rule differs from a constant multiple of the ℓ^1 subgradient. The condition is local in the center y because the potential estimates must hold near every possible unvisited site. The particular probabilities realizing the mean can vary with the past; their effect on the Green representation is a lower-order error.

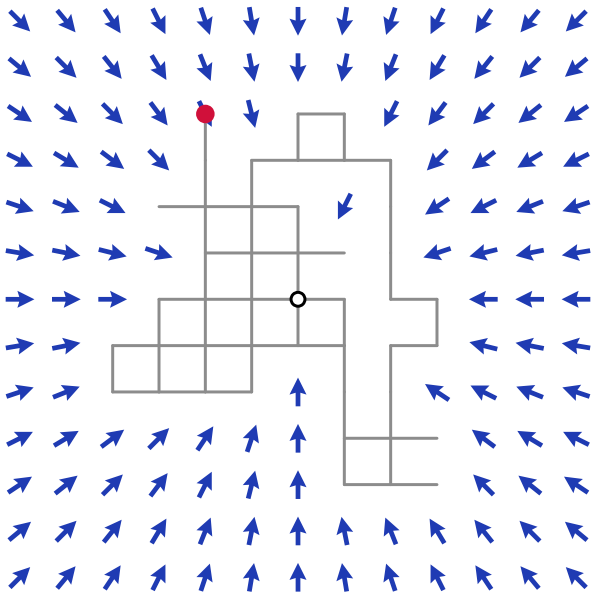


FIGURE 1. The walk after 120 steps, for $d = 2$ and $\kappa = 0.4$: its path (gray), the origin (open circle) and its position (red dot). At each site $x \neq 0$ that it has not yet left, an arrow shows the mean $-\kappa x/|x|$ of its first step from x.

1.2. **Shape, occupation, and fluctuations.** Put

$$\ell_n(x)=\sum_{j=0}^{n-1}\mathbf{1}_{\{X_j=x\}},\qquad A_n=\{x:\ell_n(x)>0\},\qquad D_n=\bigcup_{x\in A_n}\big(x+[-1/2,1/2)^d\big), \tag{1.3}$$

$$r_n=\left(\frac{(d+1)n}{2d\kappa V}\right)^{1/(d+1)},\quad b_n=\inf_{y\notin D_n}\rho(y),\quad B_n=\max_{j\le n}\rho(X_j). \tag{1.4}$$

Thus $\int 2d\kappa(r_n-\rho)_+=n$. Tildes denote constant extension of lattice functions to the unit cells. All probability estimates below concern integers $n\ge 2$. Constants may depend on d,K,κ,λ_-,A and a specified probability exponent p, but are uniform over the allowed first-departure laws.

Theorem 1.1 (Shape, occupation, and moment universality)**.** *Under (1.1)–(1.2), the following hold.*

(i) *Almost surely, for every $\varepsilon>0$ and all sufficiently large n,*

$$\{x\in\mathbb{Z}^d:\rho(x)<(1-\varepsilon)r_n\}\subset A_n\subset\{x\in\mathbb{Z}^d:\rho(x)<(1+\varepsilon)r_n\},$$

and $r_n^{-1}\max_x|\ell_n(x)-2d\kappa(r_n-\rho(x))_+|\to 0$. In particular every site is visited infinitely often.

(ii) *Define*

$$h_n=\begin{cases}\sqrt{r_n\log n}, & d=2,\\ \log n, & d\ge 3,\end{cases}\qquad q_n=h_n/r_n.$$

For every $p>0$, with probability at least $1-C_pn^{-p}$,

$$|b_n-r_n|\le C_pr_n\sqrt{q_n}, \tag{1.5}$$

$$B_n-r_n\le C_pr_nq_n^{1/(d+1)}\log n, \tag{1.6}$$

$$\max_x|\ell_n(x)-2d\kappa(r_n-\rho(x))_+|\le C_pr_nq_n^{1/(d+1)}. \tag{1.7}$$

(iii) *With h_n as in part (ii), on an event of probability at least $1-C_pn^{-p}$, simultaneously for every real $t\ge 0$,*

$$\#\{x\in A_n:\ell_n(x)\le t\}\le C_p(t+h_n)r_n^{d-1}. \tag{1.8}$$

(iv) *Let*

$$S_n=\sum_{x\in A_n}\lambda(x)\rho(x),\qquad J_K=\int_K|z|^2\,dz,\qquad \sigma_K^2=\frac{2J_K}{\kappa(d+3)}.$$

Then

$$\frac{S_n-n/(2\kappa)}{r_n^{(d+3)/2}}\Longrightarrow N(0,\sigma_K^2), \tag{1.9}$$

and, for either sign, almost surely,

$$\limsup_{n\to\infty}\frac{\pm(S_n-n/(2\kappa))}{r_n^{(d+3)/2}\sqrt{2\log\log n}}=\sigma_K. \tag{1.10}$$

For $R_{\mathrm{mom}}(n)=((d+1)S_n/(dV))^{1/(d+1)}$, the same assertions hold with $S_n-n/(2\kappa)$ replaced by $R_{\mathrm{mom}}(n)-r_n$, the exponent $(d+3)/2$ replaced by $(3-d)/2$, and σ_K replaced by $\sigma_K/(dV)$.

(v) *For some $c>0$ and all sufficiently large n, with probability at least $1-n^{-c}$,*

$$\operatorname{osc}_{|x|\le\sqrt{r_n}}\big(\ell_n(x)+2d\kappa\rho(x)\big)\ge c\begin{cases}\sqrt{r_n}\log n, & d=2,\\ \sqrt{r_n\log n}, & d\ge 3.\end{cases} \tag{1.11}$$

The oscillation is the maximum minus the minimum over lattice sites. The same orders therefore lower-bound the maximum absolute deviation from the deterministic cone in this window.

The weights in the moment statistic are essential: condition (1.1) allows an $O(r_n^d)$ change in $\sum_{A_n}\rho(x)$, which can exceed its Gaussian normalization. They disappear when $\lambda \equiv 1$.

The filled inclusion in part (i) is stronger than convergence of the outer boundary: it rules out holes throughout every smaller dilate of K. The uniform local-time statement then implies recurrence, since the cone tends to infinity at each fixed site. The occupation bound quantifies a complementary fact: few departed sites can have very small local time. The moment statistic in part (iv) is chosen so that its drift is compensated by the squared distance of the walk; this is why its fluctuations have an especially direct proof. Part (v) shows that the bulk local times fluctuate on the same orders for every convex gauge, including gauges with corners. The occupation bound uses the smaller error h_n at lightly occupied sites, rather than the larger uniform error in the bulk.

For the Euclidean case put $R_{\mathrm{in}} = b_n$, $R_{\mathrm{out}} = B_n$, and $W_n = R_{\mathrm{out}} - R_{\mathrm{in}}$. These definitions use cells for the inner radius and include X_n in the outer radius.

Theorem 1.2 (Euclidean refinement)**.** *Under the assumptions of Theorem 1.1, with $K = \{|x| \le 1\}$, all the following conclusions hold for the same class of cookie laws.*

(i) *For every $p > 0$, with probability at least $1 - C_p n^{-p}$,*

$$|R_{\mathrm{in}} - r_n| \le C_p h_n, \qquad |D_n \triangle B(0, r_n)| \le C_p r_n^{d-1} h_n, \tag{1.12}$$

$$R_{\mathrm{out}} - r_n \le C_p \begin{cases} \sqrt{r_n}(\log n)^{5/2}, & d = 2, \\ (\log n)^{d+1}, & d \ge 3, \end{cases} \tag{1.13}$$

$$\max_x |\ell_n(x) - 2d\kappa(r_n - |x|)_+| \le C_p \begin{cases} \sqrt{r_n}(\log n)^{3/2}, & d = 2, \\ \sqrt{r_n \log n}, & d \ge 3. \end{cases} \tag{1.14}$$

For each fixed $\theta < 1$, restricting the maximum to $|x| \le \theta r_n$ improves the planar bound to $C_{p,\theta}\sqrt{r_n}\log n$.

(ii) *Let G be the simple-random-walk Green function when $d \ge 3$, and put*

$$a_n = \begin{cases} \sqrt{r_n \log r_n}, & d = 2, \\ \sqrt{r_n}, & d \ge 3, \end{cases} \qquad \Sigma(y,z) = \begin{cases} 16\kappa/\pi, & d = 2, \\ 2d\kappa(2G(y-z) - \mathbf{1}_{\{y=z\}}), & d \ge 3. \end{cases}$$

For any finite set of fixed sites, the vector $a_n^{-1}(\ell_n(y) - 2d\kappa(r_n - |y|))$ converges in distribution to a centered Gaussian vector with covariance Σ. At each fixed site and for either sign, almost surely,

$$\limsup_n \frac{\pm(\ell_n(y) - 2d\kappa(r_n - |y|))}{a_n\sqrt{2\log\log n}} = \sqrt{\Sigma(y,y)}. \tag{1.15}$$

(iii) *In dimension two, for every $p > 0$ there is $c_p > 0$ such that each of*

$$\mathbb{P}\{r_n - R_{\mathrm{in}} \ge c_p\sqrt{r_n\log n}\}, \quad \mathbb{P}\{R_{\mathrm{out}} - r_n \ge c_p\sqrt{r_n\log n}\}, \quad \mathbb{P}\{W_n \ge c_p\sqrt{r_n\log n}\} \tag{1.16}$$

is at least n^{-p} for all large n. Almost surely,

$$\limsup_n \frac{r_n - R_{\mathrm{in}}}{\sqrt{r_n\log\log n}}, \quad \limsup_n \frac{R_{\mathrm{out}} - r_n}{\sqrt{r_n\log\log n}} \ge \frac{1}{\sqrt{10\pi\kappa}}, \tag{1.17}$$

$$\limsup_n \frac{W_n}{\sqrt{r_n\log\log n}} \ge \sqrt{\frac{\pi}{3\kappa}}. \tag{1.18}$$

For every $a \ge 0$,

$$\limsup_n \mathbb{P}\{W_n \le a\sqrt{r_n}\} \le 1 - \exp(-3\kappa a^2/\pi). \tag{1.19}$$

(iv) *In every dimension, for some $c > 0$ and all large n,*

$$\mathbb{P}\left\{\max_{|x|\le\sqrt{r_n}} |\ell_n(x) - 2d\kappa(r_n - |x|)| \ge c\begin{cases}\sqrt{r_n}\log n, & d=2,\\ \sqrt{r_n\log n}, & d\ge 3\end{cases}\right\} \ge 1 - n^{-c}. \tag{1.20}$$

The bounds in part (i) *hold eventually almost surely.*

We also retain the logarithmic radial lower bounds under the additional assumption of a uniform positive lower bound on all first-departure probabilities; their full tail forms are (7.17)–(7.18) below. The proof yields two additional boundary estimates. Write $R = R_{\mathrm{out}}(n) + 2\sqrt{d}$ and $E_n = D_n \setminus B(0, R_{\mathrm{in}}(n))$. With the same probability as in Theorem 1.2, uniformly over unit vectors u and $t \ge 1$,

$$|E_n \cap B(Ru, t)| \le C_p h_n t^{d-1}. \tag{1.21}$$

For lattice sites with $R_{\mathrm{in}} \le |y| \le R$, put $s = R - |y| + 1$. Then

$$\ell_n(y) \le C_p\begin{cases} h_n + \sqrt{h_n s}, & d = 2,\\ \log n + ((\log n)s^{d-1})^{1/d}, & d \ge 3.\end{cases} \tag{1.22}$$

A separate analytic consequence, Proposition 5.4, extends Nguyen's Sobolev cone inequality to every convex gauge and improves its error coefficient. Its short proof uses the same rank estimate twice; it is independent of the probabilistic estimates.

1.3. **The three models and the figures.** For the radial model of Singh [42], take $K = \{|x| \le 1\}$, $\lambda \equiv 1$, and $\kappa = \beta$. His prescribed-mean assumption is exactly (1.2). For the norm model of Bou-Rabee and Peres [9], take $\lambda \equiv 1$ and the particular law $q_x(\pm e_i) = 1/(2d) \mp \kappa\xi_i(x)/2$. Their strict parameter restriction guarantees existence and, for the logarithmic lower bounds, uniform ellipticity. Theorem 1.2 retains the Euclidean constants and exponents, and $J_{B(0,1)} = d\omega_d/(d+2)$ recovers their moment variances.

For Nguyen [37], let $K = \{\|x\|_1 \le 1\}$, $\xi(x) = \mathrm{sgn}(x)$, $\kappa = \theta/d$, and $\lambda(x) = d/m(x)$ at $x \ne 0$, where $m(x)$ is the number of its nonzero coordinates. Use his first-departure kernel. The weights differ from one only on the union of the coordinate hyperplanes, which contains at most Cs^{d-1} lattice points in any ball of radius $s \ge 1$. Thus (1.1) holds, including at $\theta = 1$. Theorem 1.1 strengthens Hausdorff convergence in probability to almost-sure filled inclusions and uniform local-time convergence. Since $|K| = 2^d/d!$, its radius is $r_n = \frac{1}{2}((d+1)!n/\theta)^{1/(d+1)}$, with the original normalization. This verifies model containment at the transition-rule level, including the exceptional coordinate hyperplanes.

Figures 2 and 3 retain the simulations from the norm model of Bou-Rabee and Peres. They illustrate the canonical kernel $q_x(\pm e_i) = 1/(2d) \mp \kappa\xi_i(x)/2$ with $\lambda \equiv 1$, a special case of the present assumptions. The same body controls both the support of the range and the level sets of the occupation profile. These pictures also show why the question of holes is distinct from the question of long, thin extensions.

1.4. **How the proof works.** The starting point is an explicit potential. If $D \subset \mathbb{R}^d$ is the set of departed cells, define

$$U_D(y) = \frac{2\kappa}{\omega_d}\int_D \nabla\rho(v)\cdot\frac{v-y}{|v-y|^d}\,dv.$$

Integration along rays from y gives the exact identity $U_{rK}(y) = 2d\kappa(r - \rho(y))_+$. Dynkin's formula for the lattice Green function gives a corresponding identity for the walk:

$$\ell_n(y) = U_{D_n}(y) - M_n^y + O(\log n),$$

where M^y is a martingale. The proof compares an unknown occupied set with the one body whose potential is already known.

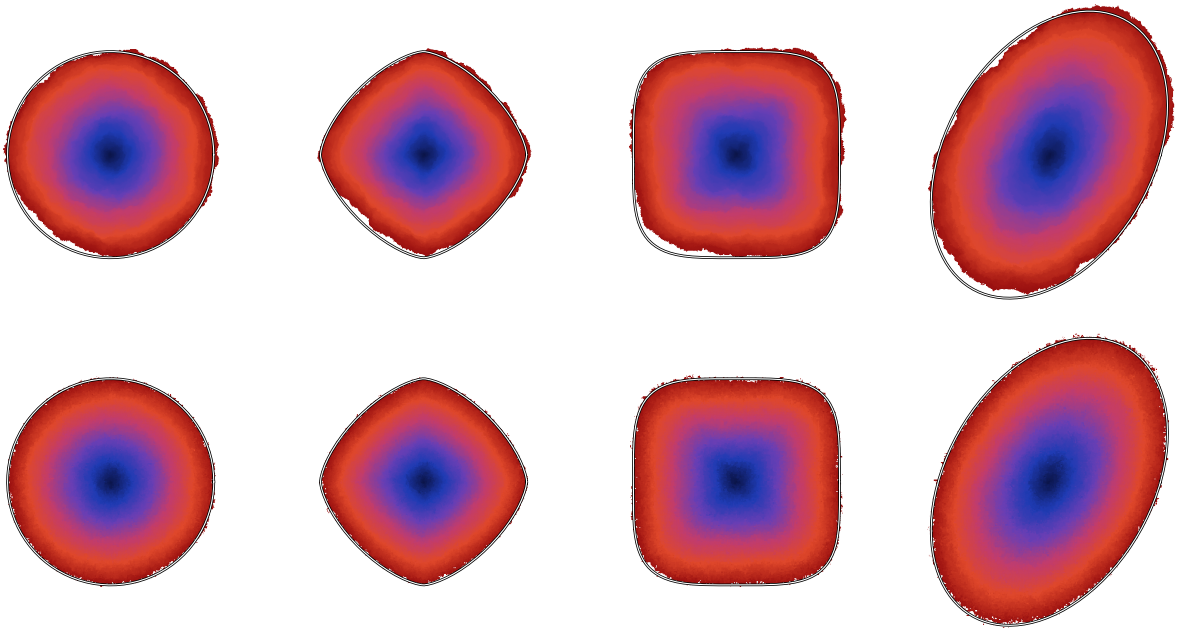


FIGURE 2. Ranges $r_n^{-1}A_n$ for the Euclidean, $\ell^{3/2}$, ℓ^4 and elliptical norms (left to right), at $n = 10^9$. Top: $d = 2$, $\kappa = 0.4$; bottom: $x_3 = 0$ sections for $d = 3$, $\kappa = 0.25$. The unit balls are outlined and the blue-to-red colors show $(t_x/n)^{\frac{1}{(d+1)}}$, with t_x the first-departure time.

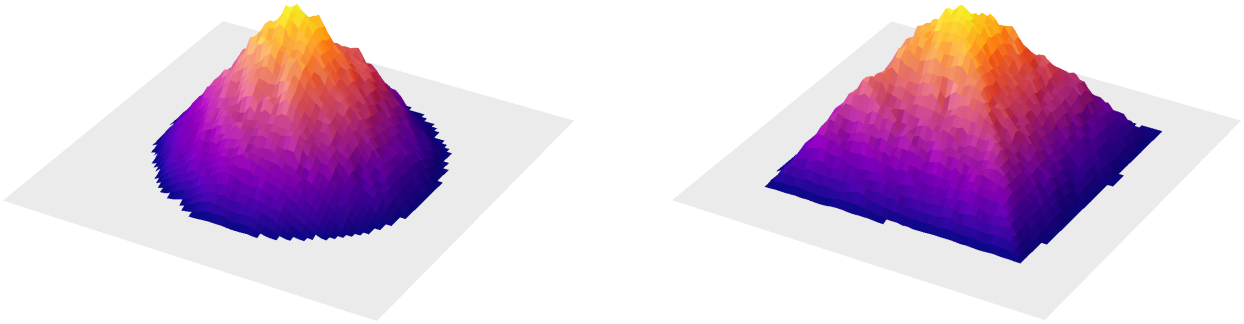


FIGURE 3. Local times $\ell_n(x)/(2d\kappa r_n)$ over x/r_n for the Euclidean norm (left) and ℓ^∞ (right): $d = 2$, $n = 10^{10}$, $\kappa = 0.4$.

First obtain the scale. Stopping at exit from a prescribed ball, as in Singh's argument, avoids estimating the range and the largest local time separately before their scale is known. If the Green martingales have not reached size R, the potential identity bounds every local time by CR. There are at most CR^d available sites, so the stopped clock is at most CR^{d+1}. The resulting variance bound makes a martingale excursion of size R unlikely. This closes the stopping argument.
Why the walk cannot make a long thin extension. An upper bound on the number of occupied sites does not by itself control the farthest point of the path. The crossing estimate uses the inward cookie at every new site to oppose such an excursion. Testing an exponential of the distance to kK, and grouping the compensator by site, lets one inward first departure pay for the at most L revisits to that site. The resulting crossing probability decays as $\exp\{-ch/(L+1)\}$ over a distance h. Thus small exterior local times imply a small exterior radius. The distance function also handles corners of K without choosing a supporting direction after seeing the path.

Why an unvisited site reveals the shape. Let bK be the largest centered gauge ball contained in D_n and let y_0 be a contact point adjacent to an empty cell. The cone potential vanishes at y_0. More importantly, convexity gives

$$\nabla\rho(v)\cdot(v-y_0)\geq\rho(v)-b\geq 0 \qquad (v\in D_n\setminus bK).$$

There is no cancellation among the excess contributions at this point. The small local time at the neighboring empty site therefore forces the excess potential to be small. This gives the conical occupation profile, and the crossing estimate excludes distant extensions. The identity $\sum_x \ell_n(x)=n$, or equivalently the quadratic martingale used below, fixes b at the deterministic radius r_n.

Why the variance estimate can be local. Apply Dynkin's formula to the square of the Green function as well as to the Green function itself. The discrete product rule expresses its edge energy in terms of the local time at the pole and a single first-departure correction per occupied site. A martingale inequality then bounds the Green noise by the local time at that same site. In particular the potential is almost nonnegative everywhere and is small at every empty site. This avoids estimating a spatial convolution of the unknown local times, and gives the smaller error needed by both the contact argument and the occupation-layer estimate.

Where the Euclidean improvement enters. For a Euclidean ball, the contact kernel has the stronger lower bound $c|v-y_0|^{2-d}/r_n$. It controls excess volume more efficiently than convexity alone. To bound the outer radius we look at an empty point beyond the enclosing sphere: nearby excess points contribute negatively there, while curvature controls the positive contribution. The positive curvature term itself is controlled by one backward radial integration of the almost-nonnegative potential. This bounds the amount of excess in a cap and hence the local times near the boundary. Figure 5 shows the two observation points. The same geometric calculation serves every dimension.

What the energy identity adds. Nguyen's calibration mechanism has a direct finite-scale form. The potential solves $\operatorname{div}(\nabla U_D+2d\kappa\mathbf{1}_D\nabla\rho)=0$. Testing this equation against a truncated potential and a cutoff of ρ bounds a nonnegative square. With zero error the square forces the cone exactly; with a small error it bounds the portion of D with small occupation. This estimate supplies an additional quantitative conclusion. The filled shape and sharp boundary bounds still use the contact and crossing arguments just described.

Why the limiting variances are universal. Once the cone is known, each typical interior site has order r_n departures, and only the first has a nonuniform law. Later departures therefore determine the leading quadratic variations. The quadratic martingale controls the weighted radial moment; Green martingales control local times. Their brackets are sums against the cone, so the martingale central limit theorem and iterated-logarithm law give the limit laws. Radius and local-time lower bounds require one additional ingredient: exponential martingales detect deviations larger than the central-limit scale. One covariance identity handles all these martingales, with only one error per first departure. The vector martingale controls planar width, and separated Green differences detect a large local-time fluctuation for every gauge. The latter lower bounds need only the coarse filled shape, while deterministic fixed-site centering needs the inner-volume estimate and qualitative outer convergence. Neither uses the final sharp outer-radius estimate.

The organization below keeps these dependencies visible. Figure 4 records the main routes through the proof.

1.5. **Related work.** Kozma [31] reported recurrence of the walk excited to the center, citing unpublished work from 2006. Dembo, Huang, and Sidoravicius [13] described the manuscript of Kozma [29] for a related walk in $d\geq 3$, with drift $-\kappa x/\|x\|_1$ at the first visit to a site having an unvisited neighbor; Dembo, Huang, and Sidoravicius [14] cited it for recurrence in every dimension. We take the name centrally excited random walk from its title. Its direction-dependent drift magnitude differs from the constant Euclidean drift in the radial special case. Benjamini, Duminil-Copin, Kozma, and Lucas [6] also recorded the predicted radius $t^{\frac{1}{(d+1)}}$.

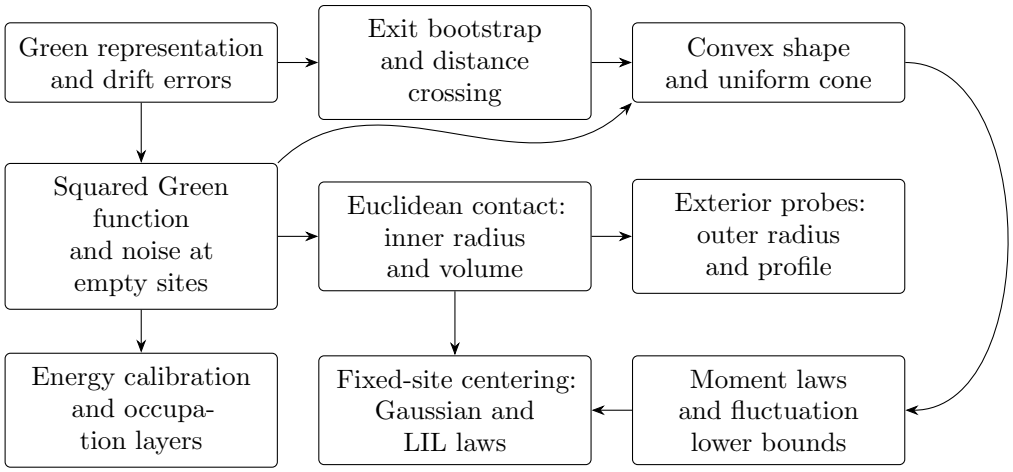


FIGURE 4. The principal estimates and their uses. The contact estimate gives both the general shape and the sharper Euclidean inner bound. The final exterior-probe argument is needed for the sharp outer-radius and uniform profile estimates. The moment laws and fluctuation lower bounds can be proved earlier, and fixed-site centering needs the inner-volume estimate but only the qualitative outer shape.

For the excited random walk with first-visit drift in a fixed direction, Benjamini and Wilson [5] proved transience in $d \geq 2$ and positive lower speed in $d \geq 4$. Kozma [27, 28] obtained positive lower speed in $d = 3$ and $d = 2$, and Bérard and Ramírez [7] proved laws of large numbers and central limit theorems. Menshikov, Popov, Ramírez, and Vachkovskaia [36] extended positive-speed results to uniformly elliptic walks with bounded jumps and uniformly positive first-visit drift in a fixed direction and zero drift at later visits. Zerner [44, 45] gave recurrence/transience criteria on $\mathbb{Z}$ and strips for walks with multiple cookies. Amir, Benjamini, and Kozma [1] proved recurrence for a walk in the upper half of $\mathbb{Z}^3$ that steps down at new sites. See Kosygina and Zerner [26] for a survey.

The radius $n^{\frac{1}{(d+1)}}$ and range size $n^{\frac{d}{(d+1)}}$ are also predicted for once-reinforced walks with strong reinforcement [4, 16], and supported by simulations [19]; related predictions for self-attracting walks appear in Ordemann et al. [38, 39]. For once-reinforced random walk on $\mathbb{Z}^d$, $d \geq 2$, Bou-Rabee and Peres [10] proved the lower bound $cn^{\frac{d}{(d+1)}}$ on the expected range for every reinforcement parameter. The matching upper bound and limit shape remain conjectural. See Pemantle [40] for reinforced processes.

Dembo et al. [15] proved an averaging principle and shape theorem for growth models motivated by origin-excited and once-reinforced walks. Their simulations with inward first steps along a coordinate of largest absolute value or along all coordinates produced squares and diamonds. The gauge theorem proves these shapes for first-departure drifts opposite to subgradients of $\|\cdot\|_\infty$ and $\|\cdot\|_1$. For related continuum growth models, Dembo and Yang [11, 12] obtained limits described by geometric flow and regularized Kardar–Parisi–Zhang equations.

Internal diffusion-limited aggregation [33] and rotor-router aggregation [34] have Euclidean ball limit shapes. The former has radius fluctuations of order at most $\log r$ in the plane and $(\log r)^{\frac{1}{2}}$ in higher dimensions [2, 3, 22, 23], and its rescaled fluctuations approach a variant of the Gaussian free field [24]. For the centrally excited walk, Theorem 1.2 gives much larger planar radius and width fluctuations; the fixed-site covariance in Theorem 1.2 is again expressed through the Green function. See Levine and Peres [35] for aggregation models.

For Eulerian walkers [41], simulations suggested a circular planar range of radius of order $t^{\frac{1}{3}}$ [25]. Florescu, Levine, and Peres [18] proved that every simple rotor walk on $\mathbb{Z}^d$ visits at least $cn^{\frac{d}{(d+1)}}$

sites in n steps. For clockwise planar rotors with independent uniform initial states, Bou-Rabee and Peres [8] proved convergence of the rescaled range to a deterministic compact convex set; whether it is a disk remains open.

1.6. **Notation and conventions.** The symbol $|\cdot|$ denotes Euclidean norm for vectors, cardinality for lattice sets, and Lebesgue measure for subsets of $\mathbb{R}^d$. We use $B(y,s)$ for the open Euclidean ball, $t_+ = \max(t,0)$, and $D \triangle E$ for symmetric difference. The filtration $\mathcal{F}_j$ contains the history through time j, and $I_j = \mathbf{1}_{\{X_j \notin A_j\}}$ records a first departure. A bracket $\langle M, N\rangle$ is predictable covariation; $\langle M\rangle = \langle M, M\rangle$. The notation M^f means the Dynkin martingale for the test function f and is linear in f.

The unit cell at x is $C_x = x + [-1/2, 1/2)^d$. A tilde denotes constant extension from lattice sites to their cells. Constants c, C may change from line to line, with the dependencies stated after (1.4); a subscript records additional dependencies. We write $a_n \asymp b_n$ when their ratio is bounded above and below by positive constants. Enlarging a constant extends an estimate valid for all sufficiently large n to every integer $n \geq 2$.

2. A common potential argument

We use the assumptions and notation of Section 1. Put $c = \min_{|u|=1} \rho(u) > 0$, $\Lambda = \max_{|u|=1} \rho(u)$, and $C_x = x + [-1/2, 1/2)^d$. Write q_x for the conditional law used at the first departure from x; this law can be random, but its prescribed mean is fixed. Every estimate below is pathwise uniform over its possible values. Set $\Delta = P - I$ for the simple-random-walk generator and $\bar{\nabla}_i f(x) = (f(x+e_i) - f(x-e_i))/2$. For a bounded measurable D, define

$$U_D(y) = \frac{2\kappa}{\omega_d} \int_D \nabla\rho(v) \cdot \frac{v-y}{|v-y|^d}\, dv. \tag{2.1}$$

Lemma 2.1 (Cone, layers, and contact)**.** *For $b \geq 0$, $U_{bK}(y) = 2d\kappa(b - \rho(y))_+$. If $E \subset \{b \leq \rho \leq b + a\}$ then $\|U_E\|_\infty \leq 2d\kappa a$; for arbitrary E, $\|U_E\|_\infty \leq C|E|^{1/d}$. Suppose $bK \subset D \subset B(0,R)$ up to null sets, $y_0 \in \partial(bK)$ lies in $\overline{D^c}$, and $E = D \setminus bK$. Put $H = U_D(y_0)$ and $I = \int_E (\rho - b)$. Then*

$$H \geq 0, \qquad I \leq CR^d H, \qquad \|U_E\|_\infty \leq CI^{1/(d+1)}. \tag{2.2}$$

Proof. For a compactly supported Lipschitz f, polar coordinates about y give $\int \nabla f(v) \cdot (v - y)|v-y|^{-d}\, dv = -d\omega_d f(y)$. Apply this to $f = (b - \rho)_+$. On each ray $y + tu$, convexity makes $t \mapsto \rho(y + tu)$ decrease and then increase. On either part, its total variation while $b \leq \rho \leq b + a$ is at most a; the two signed parts therefore give an integral between $-a$ and a. Integrating over directions proves the layer bound. Rearrangement of $|v-y|^{1-d}$ proves the volume bound. Finally $\nabla\rho(v) \cdot (v - y_0) \geq \rho(v) - \rho(y_0) = \rho(v) - b$ on E, proving the first two assertions in (2.2). Split E at $\rho = b + a$: the layer contributes at most Ca and the rest, of volume at most I/a, contributes at most $C(I/a)^{1/d}$. Take $a = I^{1/(d+1)}$; the case $I = 0$ follows directly. □

The next lemma is the bridge from the walk to the continuum potential. A first departure contributes its prescribed inward mean; its remaining covariance contributes only a second difference of the Green kernel. The only geometric approximation needed at a corner of K is a one-dimensional telescoping estimate.

Lemma 2.2 (Discrete potential representation)**.** *Let g be the potential kernel in dimension two and minus the Green function in dimensions at least three, normalized by $\Delta g = \mathbf{1}_{\{0\}}$, where $\Delta = P - I$ for simple random walk. For any lattice function f, write M^f for its Dynkin martingale; this notation is linear in f. Set $g_y(x) = g(x-y)$ and $M^y = M^{g_y}$. The Green martingales have bounded increments. Whenever $\max_{j \leq t} |X_j| \leq R$, $R \geq 2$, simultaneously for all $y \in \mathbb{Z}^d$,*

$$\ell_t(y) = U_{D_t}(y) - M_t^y + O(\log R), \qquad \langle M^y\rangle_t \leq CW_t(y), \quad W_t(y) = \sum_x \ell_t(x)(1 + |x-y|)^{2-2d}. \tag{2.3}$$

Moreover, for $|z-y| \le \sqrt{d}$, $|U_{D_t}(z) - U_{D_t}(y)| \le C \log R$.

Proof. Standard Green estimates [32] give, for a coordinate unit vector e,

$$\bar{\nabla} g(z) = \frac{2}{\omega_d} \frac{z}{|z|^d} + O((1+|z|)^{-d}) \quad (z \neq 0),$$
$$|g(z+e) - g(z)| \le C(1+|z|)^{1-d},$$
$$|g(z+e_i) + g(z-e_i) - 2g(z)| \le C(1+|z|)^{-d}.$$

At the first departure from x, put $s_i(x) = q_x(e_i) + q_x(-e_i)$. The exact decomposition is

$$(Q_x - P) g_y(x) = -\kappa \lambda(x) \xi(x) \cdot \bar{\nabla} g(x-y) + \frac{1}{2} \sum_i (s_i(x) - 1/d)[g(x+e_i-y) + g(x-e_i-y) - 2g(x-y)].$$

Thus the arbitrary first-departure covariance has coefficients bounded by one and appears only in the second line. Dynkin's formula, grouping first departures by site, reads

$$\ell_t(y) = g(X_t - y) - g(-y) - M_t^y - \sum_{x \in A_t} (Q_x - P) g_y(x).$$

The endpoint term is $O(\log R)$ uniformly in y: in dimension two use $g(z) = (2/\pi)\log(1+|z|) + O(1)$, and in higher dimensions g is bounded. Summing the second differences over the occupied sites costs $O(\log R)$. The same estimate replaces $\bar{\nabla} g$ by its leading term; define that term to be zero at the origin.

The defects also cost only $O(\log R)$. Indeed, the hypothesis on λ and a decomposition into dyadic annuli give

$$\sup_y \sum_{x \in B(0, CR)} |\lambda(x) - 1| (1 + |x-y|)^{1-d} \le C \log R. \tag{2.4}$$

For $|y| \le 2CR$, each annulus contributes at most a constant. For $|y| > 2CR$, the total defect mass is at most CR^{d-1} and the kernel is at most $C|y|^{1-d}$, so the bound is constant. The estimate applies to any subset of the defect sites.

Here is an elementary substitute for a smoothness assumption on ρ. Define its nonnegative lattice curvature by

$$\eta(x) = \sum_{i=1}^{d} [\rho(x + de_i) + \rho(x - de_i) - 2\rho(x)].$$

Then, for every choice $\xi(x) \in \partial\rho(x)$, including $\xi(0) = 0$,

$$\int_{C_x} |\nabla \rho(v) - \xi(x)| \, dv \le C\eta(x), \qquad \sum_{x \in B(y,s) \cap \mathbb{Z}^d} \eta(x) \le C s^{d-1} \quad (s \ge 1). \tag{2.5}$$

To prove the first assertion, subtract the supporting plane and put $h(v) = \rho(v) - \rho(x) - \xi(x) \cdot (v - x)$. This is convex, nonnegative, and zero at x. For $v \in C_x$, the points $v \pm e_i$ lie in the convex hull of $x \pm de_j$, because their ℓ^1 distance from x is at most $d/2 + 1 \le d$. Their h-values are therefore at most $\eta(x)$. At every differentiability point, convexity on a coordinate line gives

$$|\partial_i h(v)| \le h(v + e_i) + h(v - e_i) \le 2\eta(x),$$

which proves the cell estimate. For the second assertion, sum each coordinate second difference along the lattice lines meeting the ball. The points on each line form an interval, and the sum telescopes to at most $2d$ first differences at its endpoints. Each first difference has absolute value at most $d\Lambda$, and there are $O(s^{d-1})$ such lines. This proves the claimed bound.

Write $F_y(v) = (v-y)|v-y|^{-d}$. Cells at bounded distance from y have a uniformly bounded total contribution. On the other cells, $|F_y(v) - F_y(x)| \le C(1+|x-y|)^{-d}$, whose sum is $O(\log R)$. The remaining error, from replacing $\xi(x)$ by $\nabla\rho(v)$, is at most

$$C \sum_{x \in B(0,CR)} \eta(x)(1+|x-y|)^{1-d} = O(\log R).$$

This follows from (2.5) by exactly the near-pole and far-pole annulus argument used for (2.4). We have now obtained U_{D_t} and the first identity in (2.3), uniformly over all poles.

Finally a martingale increment is a centered nearest-neighbor increment of g_y. Its conditional variance is at most $C(1+|X_j-y|)^{2-2d}$, for every allowed conditional step law. Summing gives the bracket bound. To obtain the cell modulus, integrate the kernel directly within distance $2\sqrt{d}$ of either pole and use $|\nabla F_y(v)| \le C|v-y|^{-d}$ elsewhere. The same near-pole and far-pole split gives $C\log R$. □

Lemma 2.3 (Exit bootstrap). *Put $\tau_R = \inf\{t : \rho(X_t) \ge R\}$. For every $p > 0$, outside an event of probability CR^{-p}, simultaneously for $t \le \tau_R$,*

$$\max_x \ell_t(x) \le CR, \qquad \tau_R \le CR^{d+1}, \qquad \sup_{y \in \mathbb{Z}^d} |M_t^y| \le C\sqrt{R}\log R. \tag{2.6}$$

Consequently $\|\widetilde{\ell}_t - U_{D_t}\|_\infty \le C\sqrt{R}\log R$.

Proof. The point of the extra stop below is to obtain a deterministic clock bound before applying a martingale inequality. No prior estimate of τ_R, or even of its finiteness, is needed. For $t \le \tau_R$, every departed site lies in $B(0, C_1R)$. The exact Dynkin identity and the Green increment bound give

$$\ell_t(y) \le |M_t^y| + CR$$

uniformly in y: the sum of $(1+|x-y|)^{1-d}$ over a ball of radius C_1R is at most CR, and the endpoint costs at most $C\log R$. Choose C_2 large and set

$$\mathcal{Y}_R = \{y \in \mathbb{Z}^d : |y| \le C_2R^2\}, \qquad \gamma_R = \inf\{t : \max_{y \in \mathcal{Y}_R} |M_t^y| \ge R\}, \qquad \sigma_R = \tau_R \wedge \gamma_R.$$

At every integer $t < \sigma_R$, all occupied sites belong to $\mathcal{Y}_R$ and have local time at most CR. Since there are at most CR^d such sites, $t = \sum_x \ell_t(x) \le CR^{d+1}$. Thus $\sigma_R \le CR^{d+1} + 1$ deterministically, even if τ_R was initially allowed to be infinite. The bounded martingale overshoot and the single final departure preserve $\max_x \ell_{\sigma_R}(x) \le CR$. Consequently

$$\langle M^y \rangle_{\sigma_R} \le CR \sum_{x \in B(0,C_1R)} (1+|x-y|)^{2-2d} \le CR\log R.$$

Freedman's maximal inequality [20, Theorem 1.6], applied to the stopped martingales, and a union bound over $|\mathcal{Y}_R| = O(R^{2d})$ poles show that

$$\max_{y \in \mathcal{Y}_R} \max_{t \le \sigma_R} |M_t^y| \le C_p\sqrt{R}\log R$$

except with probability CR^{-p}. For sufficiently large R this is less than R. Hence $\gamma_R \le \tau_R$ is impossible on that event, and $\sigma_R = \tau_R$ gives both the clock and the local-time bounds. Bounded values of R are absorbed by increasing the constants.

There is no infinite union of exceptional events hidden in the stated supremum. For $|y| > C_2R^2$, the pole is unvisited before exit and the exact Dynkin identity gives, pathwise for $t \le \tau_R$,

$$|M_t^y| \le C(R + |A_t|)|y|^{1-d} \le CR^d|y|^{1-d} \le C.$$

Here the endpoint uses the Green increment bound along a lattice path of length $O(R)$, and each first departure contributes at most $C|y|^{1-d}$. This proves the assertion for every pole. The representation and cell modulus complete the proof. □

The next estimate converts small exterior local times into a spatial bound. We use the distance to the whole convex body, so the argument needs no choice or discretization of supporting directions.

Lemma 2.4 (Distance crossing)**.** *For every $p > 0$ and integer $N \geq 2$, outside an event of probability CN^{-p}, simultaneously for $t \leq N$ and every real $k \geq 0$,*

$$\max_{j \leq t} \operatorname{dist}(X_j, kK) \leq C \left(1 + \max_{\rho(x) \geq k} \ell_t(x)\right) \log N. \tag{2.7}$$

Proof. Fix $k \geq 0$ and write $D(x) = \operatorname{dist}(x, kK)$. If $D(x) > 2$, let $\pi(x)$ be the Euclidean projection onto kK and $\nu(x) = (x - \pi(x))/D(x)$. For any nearest-neighbor step e,

$$D(x + e) \leq |x + e - \pi(x)| \leq D(x) + \nu(x) \cdot e + C/D(x).$$

Moreover, the subgradient inequality and the competitor $kx/\rho(x) \in kK$ imply

$$\xi(x) \cdot \nu(x) \geq \frac{\rho(x) - k}{D(x)} \geq c.$$

For $f(x) = \exp(\theta D(x))$, $0 < \theta \leq 1$, use the preceding distance bound and $|D(x + e) - D(x)| \leq 1$. If I_j indicates a first departure, then, whenever $D(X_j) > 2$,

$$\mathbb{E}[f(X_{j+1}) - f(X_j) \mid \mathcal{F}_j] \leq \theta f(X_j) \left[C \left(\theta + \frac{1}{D(X_j)}\right) - \kappa \lambda_- c I_j\right]. \tag{2.8}$$

This uses only the conditional mean; all second-order terms are bounded by the unit step size.

Fix an integer $L \geq 0$ and $H > 0$. Stop at time N, at the first arrival with $D(X_j) \geq H$, or as soon as some site with $\rho(x) \geq k$ has local time $L + 1$, whichever comes first; call this time T. Every exterior site has at most $L + 1$ departures before T. Choose $\theta = c_0/(L + 1)$ and a fixed $A_0 > 2$, with c_0 sufficiently small and A_0 sufficiently large that

$$C(c_0 + A_0^{-1}) \leq \kappa \lambda_- c.$$

For a departed site x with $D(x) > A_0(L+1)$, sum the right-hand side of (2.8) over all its departures before T. Its cookie occurs exactly once, so that sum is at most

$$\theta f(x) \left[(L + 1)C \left(\theta + D(x)^{-1}\right) - \kappa \lambda_- c\right] \leq 0.$$

In the remaining tube, $f(X_j)$ and $f(X_{j+1})$ are bounded by the fixed constant $\exp(c_0(A_0 + 1))$, so each compensator term is bounded above by a fixed constant. There are at most N terms. Optional stopping of the bounded-time Dynkin martingale therefore gives $\mathbb{E} f(X_T) \leq 1 + CN$. In particular,

$$\mathbb{P}\left\{ \begin{array}{c} \text{for some } t \leq N, \quad \max_{j \leq t} D(X_j) \geq H, \\ \max_{\rho(x) \geq k} \ell_t(x) \leq L \end{array} \right\} \leq CN \exp\{-c_0 H/(L + 1)\}. \tag{2.9}$$

Indeed, on this event the first arrival at distance H precedes the local-time stop, so $f(X_T) \geq e^{\theta H}$. The argument has not conditioned on the event whose probability is being estimated.

Apply (2.9) with $H = C_p(L + 1) \log N$, and take a union bound over the integers $0 \leq L \leq N$ and $0 \leq k \leq \lceil \Lambda N \rceil + 1$. There are $O(N^2)$ choices; increasing C_p makes their total failure probability at most CN^{-p}. At every time $t \leq N$, substitute $L = \max_{\rho(x) \geq k} \ell_t(x)$ to get (2.7) for these integer levels. Larger levels contain the whole path, since $\rho(X_j) \leq \Lambda N$. Finally, for real $k \geq 0$, set $m = \lceil k \rceil$. Radial scaling inside $K \subset B(0, 1/c)$ gives

$$D_k(x) \leq D_m(x) + 1/c, \qquad \max_{\rho(x) \geq m} \ell_t(x) \leq \max_{\rho(x) \geq k} \ell_t(x).$$

The integer estimate, with the fixed additive constant absorbed, proves the result for all real levels simultaneously. □

Proposition 2.5 (Coarse scale, with polynomial tails)**.** *For every $p > 0$ there are $c_p, C_p > 0$ such that, with probability at least $1 - C_p R^{-p}$,*

$$c_p R^{d+1} \le \tau_R \le C_p R^{d+1}. \tag{2.10}$$

Consequently, with $r_n = ((d+1)n/(2d\kappa|K|))^{1/(d+1)}$, with probability at least $1 - C_p n^{-p}$,

$$\begin{gathered} \max_{j\le n} |X_j| \le C_p r_n, \qquad c_p r_n \le \max_x \ell_n(x) \le C_p r_n, \qquad c_p r_n^d \le |A_n| \le C_p r_n^d, \\ c_p r_n K \subset D_n, \qquad \min_{\substack{x\in\mathbb{Z}^d \\ \rho(x)\le c_p r_n}} \ell_n(x) \ge c_p r_n. \end{gathered} \tag{2.11}$$

Proof. Work on the events of the preceding two lemmas with a deterministic horizon $N = \lceil CR^{d+1} \rceil$. At time τ_R, let $b = \inf_{y\notin D_{\tau_R}} \rho(y)$, and choose a contact point $y_0 \in \partial(bK) \cap \overline{D^c_{\tau_R}}$ in the closure of an unoccupied cell. Continuity and the cell estimate imply $0 \le H = U_{D_{\tau_R}}(y_0) \le \delta_R = C\sqrt{R}\log R$. Lemma 2.1 therefore gives

$$\|\widetilde{\ell}_{\tau_R} - 2d\kappa(b-\rho)_+\|_\infty \le \delta_R + C(R^d\delta_R)^{1/(d+1)} =: \varepsilon_R = o(R/\log R).$$

Use (2.7) with $k = b$; at sites outside bK the local times are at most ε_R. Since $\rho(X_{\tau_R}) \ge R$ and ρ is Λ-Lipschitz,

$$R - b \le C + C(\varepsilon_R + 1)\log R = o(R).$$

The cone is supported on $\{\rho < b\} \subset D_{\tau_R}$. Integrating its approximation over $D_{\tau_R} \subset B(0, CR)$ gives

$$\tau_R = \frac{2d\kappa|K|}{d+1} b^{d+1} + O(R^d \varepsilon_R) \asymp R^{d+1},$$

proving (2.10). To obtain (2.11), choose $R = \lceil Cn^{1/(d+1)} \rceil$ so that the lower clock bound exceeds n, and apply the exit event with exponent $(d+1)(p+1)$. Then $n < \tau_R$, so the spatial and local-time upper bounds follow. For the inner bounds, also choose $R' = \lfloor cn^{1/(d+1)} \rfloor$ with c sufficiently small that the upper clock bound gives $\tau_{R'} \le n$. The contact argument just proved gives $b_{\tau_{R'}} \ge 3R'/4$ for large n, and its cone approximation has error $o(R')$. Hence every lattice site with $\rho(x) \le R'/2$ has local time at least $c'R'$ at $\tau_{R'}$. These local times only increase before time n. The occupied cells therefore contain $(R'/4)K$, after absorbing their fixed diameter. Reducing c_p gives both final assertions in (2.11), as well as the lower bounds for maximum local time and volume. A union bound for the two exit events retains probability $1 - C_p n^{-p}$. □

3. Local noise from a squared Green function

Fix $p > 0$, write $r = r_n$ and $a = \log n$, and let g be the planar potential kernel or minus the Green function in dimensions at least three, normalized by $\Delta g = \mathbf{1}_{\{0\}}$. On the coarse event of Proposition 2.5,

$$D_n \subset B(0, Cr), \qquad |A_n| \le Cr^d, \qquad \max_x \ell_n(x) \le Cr. \tag{3.1}$$

The useful noise estimate is local: when a site has little occupation, the martingale in its Green representation has little variation. This follows by applying Dynkin's formula a second time, to a square.

Lemma 3.1 (Squared-Green estimate)**.** *There is an event of probability at least $1 - Cn^{-p}$ on which (3.1) holds and, simultaneously for all lattice sites $|y| \le 3n$,*

$$|\ell_n(y) - U_{D_n}(y)| \le \begin{cases} C\big(\sqrt{ra} + a\sqrt{\ell_n(y)}\big), & d = 2, \\ C\big(a + \sqrt{a\ell_n(y)}\big), & d \ge 3. \end{cases} \tag{3.2}$$

In particular, writing

$$h = \begin{cases} \sqrt{ra}, & d = 2, \\ a, & d \geq 3, \end{cases} \qquad q = h/r, \tag{3.3}$$

we have, after constant extension of the local times to cells,

$$U_{D_n} \geq \tfrac{1}{2}\widetilde{\ell}_n - Ch, \qquad |U_{D_n}| \leq Ch \quad \text{on } D_n^c. \tag{3.4}$$

Proof. For a lattice function f, let M^f denote its Dynkin martingale. Define the edge energy

$$e(z) = \frac{1}{2d} \sum_{|v|=1} (g(z+v) - g(z))^2, \qquad \mathcal{E}_n(y) = \sum_x \ell_n(x) e(x-y).$$

Since every conditional step law is supported on the $2d$ neighbors, $\langle M^{g_y} \rangle_n \leq 2d\mathcal{E}_n(y)$. For any constant c, the exact product identity is

$$\Delta(g-c)^2 = 2(g(0)-c)\mathbf{1}_{\{0\}} + e. \tag{3.5}$$

Let $f_y(x) = (g(x-y)-c)^2$ and $N^y = M^{f_y}$. Applying Dynkin's formula and collecting the first-departure corrections gives

$$\begin{aligned} \mathcal{E}_n(y) = {} & 2(c - g(0))\ell_n(y) + f_y(X_n) - f_y(0) - N_n^y \\ & - \sum_{x \in A_n} \sum_{|v|=1} \left(q_x(v) - \frac{1}{2d} \right) (f_y(x+v) - f_y(x)). \end{aligned}$$

Each site contributes to the last sum only once.

If $d \geq 3$, take $c = 0$. The function $g = -G$ is bounded, and the Green gradient estimate gives

$$\max_{|v|=1} |g(z+v)^2 - g(z)^2| \leq C(1+|z|)^{3-2d}.$$

The sum of these bounds over any Cr^d sites is $O(\log r)$ when $d = 3$ and $O(1)$ when $d \geq 4$, uniformly in the pole y. The endpoint terms are bounded. Also $\langle N^y \rangle_n \leq C\mathcal{E}_n(y)$ and N^y has bounded increments, since $|\Delta_v(g^2)| \leq C|\Delta_v g|$. Freedman's inequality and the displayed identity therefore give

$$\mathcal{E}_n(y) \leq C\ell_n(y) + Ca + C\sqrt{a\mathcal{E}_n(y)}, \qquad \text{hence} \qquad \mathcal{E}_n(y) \leq C(\ell_n(y) + a).$$

In the plane choose a deterministic $R_0 \asymp r$, sufficiently large, and $c = (2/\pi)\log R_0$. The Green estimates imply

$$\max_{|v|=1} |(g(z+v)-c)^2 - (g(z)-c)^2| \leq C \frac{1 + |\log((1+|z|)/R_0)|}{1+|z|}.$$

The radial majorant decreases with $|z|$. Its sum over the Cr^2 lattice points nearest to the pole is at most Cr: in polar coordinates the relevant integral is bounded by $C\int_1^{Cr}(1+|\log(s/R_0)|)\,ds \leq Cr$. Thus the first-departure correction is $O(r)$ uniformly even for poles far from the range. Equivalently, for $|y| \gg r$ its direct bound is $Cr^2(1+\log(|y|/r))/|y| \leq Cr$. For all poles $|y| \leq 3n$ and times up to n, the endpoint terms are $O(a^2)$, the increments of N^y are $O(a)$, and $\langle N^y \rangle_n \leq Ca^2\mathcal{E}_n(y)$. Consequently

$$\mathcal{E}_n(y) \leq Ca\ell_n(y) + Cr + Ca^2 + Ca\sqrt{a\mathcal{E}_n(y)} \leq \tfrac{1}{2}\mathcal{E}_n(y) + C(a\ell_n(y) + r + a^3).$$

Since $a^3 = o(r)$, this gives $\mathcal{E}_n(y) \leq C(a\ell_n(y) + r)$.

The random-bracket forms of Freedman used here follow by splitting $\mathcal{E}_n(y) \leq Cn$ into dyadic bins. Their $O(\log n)$ cost and a union bound over the $O(n^d)$ poles are absorbed by increasing the constant multiplying a. Applying Freedman to M^{g_y}, whose increments are bounded and whose bracket is at most $2d\mathcal{E}_n(y)$, and then using Lemma 2.2, proves (3.2). Young's inequality absorbs half the local time in its lower bound for U_{D_n}. In the plane the additional $O(a^2)$ is absorbed by

h. At an empty site set $\ell_n(y) = 0$. The cell modulus extends these conclusions to the continuum; outside the pole region use the decay of the potential. This proves (3.4). □

The same estimate and $\max \ell_n \le Cr$ give the familiar two-sided uniform error

$$\|\widetilde{\ell}_n - U_{D_n}\|_\infty \le \delta, \qquad \delta = \begin{cases} C\sqrt{r}\,a, & d = 2, \\ C\sqrt{ra}, & d \ge 3. \end{cases} \tag{3.6}$$

Define

$$b = \inf_{y \notin D_n} \rho(y), \quad E = D_n \setminus bK, \quad H = U_{D_n}(y_0), \quad I = \int_E (\rho(v) - b)\, dv,$$

where $y_0 \in \partial(bK)$ lies in the closure of an empty cell C_z. Convexity gives $H = U_E(y_0) \ge 0$.

Lemma 3.2 (Noise at the first hole)**.** *On the preceding event, $H \le Ch$.*

Proof. This is (3.4) at y_0, by continuity from the empty cell. No geometric estimate of a martingale bracket is needed. □

Rates for every convex body. Lemma 2.1 and the contact bound give $I \le Cr^d h = Cr^{d+1}q$. Split E into the layer $b \le \rho < b + s$, whose volume is at most $Cr^{d-1}s$, and the remainder, whose volume is at most I/s. With $s = r\sqrt{q}$ this proves

$$|E| \le Cr^d\sqrt{q}, \qquad \|U_E\|_\infty \le Crq^{1/(d+1)}. \tag{3.7}$$

The quadratic martingale is

$$Q_n = |X_n|^2 - n + 2\kappa \sum_{x \in A_n} \lambda(x)\rho(x).$$

Stopping on leaving $B(0, Cr)$ and applying Azuma gives $|Q_n| \le Cr\sqrt{na}$. The sparse-weight condition and cell quadrature each cost $O(r^d)$ in replacing the weighted sum by $\int_{D_n} \rho$. Since $\int_{bK} \rho = d|K|b^{d+1}/(d+1)$ and $n = 2d\kappa|K|r^{d+1}/(d+1)$, we obtain

$$\left|1 - (b/r)^{d+1}\right| \le C\left(r^{(1-d)/2}\sqrt{a} + r^{-1} + |E|r^{-d}\right). \tag{3.8}$$

The first two terms are at most $C\sqrt{q}$; hence

$$|b - r| \le Cr\sqrt{q}, \qquad \max_x |\ell_n(x) - 2d\kappa(r - \rho(x))_+| \le Crq^{1/(d+1)}. \tag{3.9}$$

Here we used the cone identity, (3.6) and (3.7); their errors are bounded by the right side. Apply the distance crossing estimate (2.7) with $k = r$. Outside rK the local time is at most $Crq^{1/(d+1)}$. Since ρ is Lipschitz, it follows that

$$\max_{j \le n} \rho(X_j) \le r + Crq^{1/(d+1)}a. \tag{3.10}$$

These prove Theorem 1.1(ii). Taking $p > 1$ and applying Borel–Cantelli gives the filled inclusions and uniform cone law in part (i); at each fixed site the cone grows linearly in r_n, proving recurrence.

4. Euclidean geometry and exterior probes

Now let $\rho(v) = |v|$. We retain r, a, b, E, h, q above, so $b = R_{\text{in}}(n)$. For $|v| \ge b = |y_0|$ the exact identity

$$2(|v|^2 - v \cdot y_0) = |v - y_0|^2 + |v|^2 - b^2$$

gives

$$u_v \cdot \frac{v - y_0}{|v - y_0|^d} \ge \frac{|v - y_0|^{2-d}}{2|v|} \ge cr^{1-d} \qquad (v \in E).$$

Thus the contact estimate improves the volume and inner radius bounds to

$$|E| \le Cr^{d-1}h = Cr^d q, \quad |b-r| \le Crq, \quad |D_n \triangle B(0,r)| \le Cr^d q. \tag{4.1}$$

The radius assertion follows from (3.8) since its first two terms are at most Cq.

Write U_E^+ for the radial potential with its negative integrand replaced by zero, and put

$$L = 1 + \max_{|x|\ge b} \ell_n(x), \qquad w = R_{\mathrm{out}}(n) - b + 2\sqrt{d}, \qquad R = b + w.$$

Then $w \ge 1$ and $E \subset \{b \le |v| \le R\}$. The one-sided squared-Green bound gives directly, in every dimension,

$$\ell_n(y) \le C\big((b-|y|)_+ + \|U_E\|_\infty + h\big), \qquad \ell_n(y) \le C(U_E^+(y) + h) \quad (|y| \ge b). \tag{4.2}$$

Direct control of the curvature term. In every dimension,

$$\sup_y \int_E |v-y|^{2-d}\, dv \le Crh. \tag{4.3}$$

For $d = 2$ this is the volume bound. For $d \ge 3$, put $F(y) = \int_E |v-y|^{2-d}/|v|\, dv$. The coarse bounds give $b \asymp r$ and $R \asymp r$, so $F(2Ru) \le Cr^{1-d}|E| \le Ca$ for every unit vector u. Direct differentiation yields

$$\frac{d}{dt}\big(t^{d-2}F(tu)\big) = \frac{\omega_d(d-2)}{2\kappa} t^{d-3} U_E(tu). \tag{4.4}$$

For $t \ge b$ the cone vanishes, so $U_E(tu) = U_{D_n}(tu) \ge -Ca$ by (3.4). Integrating backward from $2R$ gives, for $b \le t \le 2R$,

$$t^{d-2}F(tu) \le (2R)^{d-2}F(2Ru) + Ca\int_t^{2R} s^{d-3}\, ds \le Cr^{d-2}a.$$

Thus $F \le Ca$ throughout this annulus. The same bound holds inside $B(0,b)$ and outside $B(0,2R)$ by the maximum principle, because F is harmonic there and tends to zero at infinity. Finally $|v| \le Cr$ on E, proving (4.3).

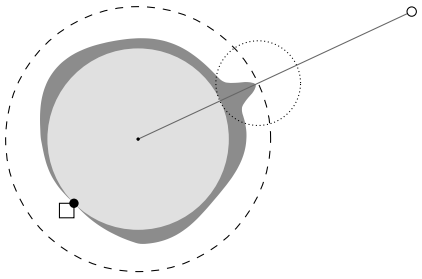


FIGURE 5. The light disk is $B(0,b)$, the largest ball about the origin in D_n, the dark region is $E = D_n \setminus B(0,b)$, and the dashed circle, of radius $b+w$, contains D_n. The square is a cell disjoint from D_n whose closure contains the point y_0 (black dot) of the circle $\{|v| = b\}$, and every point of E contributes nonnegatively to $U_E(y_0)$. The segment joins the origin (small dot) to $y_t = (b+w+4t)u$ (open dot) for a unit vector u, and the points of E in the dotted ball $B((b+w)u, t)$ contribute negatively to $U_E(y_t)$.

One exterior estimate in all dimensions. For $|y| \ge R$, the Euclidean identity and (4.3) give $U_E^+(y) \le Cr^{-1}\int_E |v-y|^{2-d}\,dv \le Ch$.

We now use the same exterior probe in every dimension. Given a unit vector u and $\sqrt{d} \le t \le r/4$, put $y_t = (R+4t)u$. This point lies outside D_n, and the cone vanishes there. Therefore (3.4) gives the stronger estimate

$$|U_E(y_t)| = |U_{D_n}(y_t)| \le Ch. \tag{4.5}$$

For $v \in E \cap B(Ru,t)$ we have $u_v \cdot u \ge 1 - t/R$ and

$$u_v \cdot (v - y_t) \le R - (R+4t)(1-t/R) \le -t, \qquad |v - y_t| \le 5t,$$

because $t \le R/2$ for large n. Such points contribute at most $-ct^{1-d}$ each, whereas all positive contributions total at most Ch. Hence (4.5) implies

$$|E \cap B(Ru,t)| \le Cht^{d-1} \qquad (|u| = 1,\ t \ge 1). \tag{4.6}$$

For $t > r/4$ use $|E| \le Cr^{d-1}h$; for $1 \le t < \sqrt{d}$ use the ball-volume bound. Thus the estimate holds down to a fixed spatial scale, independently of the shell width.

Lemma 4.1 (Cap volume controls the boundary profile)**.** *For every y with $b \le |y| \le R$, put $s = R - |y| + 1$. Then*

$$U_E^+(y) \le C(hs^{d-1})^{1/d} + Ch.$$

The same bound holds for $\ell_n(y)$ at lattice sites.

Proof. For $t \ge s$, $B(y,t) \subset B(Ru_y, 2t)$, so (4.6) gives $|E \cap B(y,t)| \le Cht^{d-1}$. The part $|v-y| \le s$ has volume at most Chs^{d-1} and therefore potential at most $C(hs^{d-1})^{1/d}$. For the rest, $|v|^2 - |y|^2 \le 2Rs$ and $|v| \ge b \asymp r$ give

$$\left[u_v \cdot \frac{v-y}{|v-y|^d}\right]_+ \le \frac{Cs}{|v-y|^d} + \frac{C}{r|v-y|^{d-2}}.$$

The cap bound makes the first integral at most $Cs\int_s^\infty ht^{d-1}t^{-d-1}\,dt \le Ch$; (4.3) bounds the second by Ch. The exterior local-time bound proves the last assertion. □

Since $s \le w + 1 \le 2w$, the lemma and distance crossing with $k = b$ give

$$L \le C\big((hw^{d-1})^{1/d} + h\big), \qquad w \le CLa \le \tfrac{1}{2}w + Cha^d. \tag{4.7}$$

Thus $w \le Cha^d$ in every dimension. In the plane the lemma yields

$$w \le C\sqrt{r}\,a^{5/2}, \qquad \sup_{b \le |y| \le R} U_E^+(y) \le C\sqrt{r}\,a^{3/2}. \tag{4.8}$$

For $d \ge 3$, $h = a$, and the same inequality immediately gives

$$w \le Ca^{d+1}. \tag{4.9}$$

The all-scale cap estimates are already contained in (4.6). Lemma 4.1 now also gives

$$\ell_n(y) \le \begin{cases} C\big(h + \sqrt{hs}\big), & d = 2, \\ C\big(a + (as^{d-1})^{1/d}\big), & d \ge 3, \end{cases} \qquad b \le |y| \le R, \quad s = R - |y| + 1. \tag{4.10}$$

In particular, for $d \ge 3$ the outermost radial layer of width $C_0 \log n$ has local times $O(\log n)$. If this layer reaches inside $B(0,b)$, then $w = O(a)$. The continuum boundary bound from the lemma and the maximum principle give $U_E \le Ca$ inside that ball. The first inequality in (3.4), together with the cone depth $O(a)$, gives the same local-time bound there.

Uniform and interior local times. For $d \geq 3$, the layer bound and (4.9) give $\|U_E\|_\infty \leq Ca^{d+1}$. Together with the cone identity, (3.6) and (4.1), this gives

$$\max_x |\ell_n(x) - 2d\kappa(r - |x|)_+| \leq C\sqrt{ra}.$$

In the plane U_E is nonnegative and harmonic on $B(0, b)$, and continuous on its closure. The maximum principle and (4.8) therefore bound it throughout that ball by $C\sqrt{r}\, a^{3/2}$. On the outer annulus use (4.2); beyond the range both the local time and, except for a radial error $C|b - r|$, the target cone vanish. Consequently the uniform planar error is $C\sqrt{r}\, a^{3/2}$.

Finally, for any fixed $\theta < 1$ and $|y| \leq \theta r$, the excess set is at distance at least $c_\theta r$ for large n. Thus

$$|U_{D_n}(y) - 2d\kappa(r - |y|)| \leq C_\theta(r^{1-d}|E| + |b - r|) \leq C_\theta rq. \tag{4.11}$$

The corresponding local-time error is at most $C_\theta\sqrt{r}\, a$ in the plane and $C_\theta\sqrt{ra}$ in higher dimensions. All assertions hold simultaneously with probability $1 - Cn^{-p}$ and, by Borel–Cantelli, eventually almost surely. The all-scale cap bounds (4.6) and boundary profile (4.10) are additional quantitative conclusions of the same proof.

5. Calibration and small occupation layers

Lemma 5.1 (Finite-scale calibration)**.** *Let $a = 2d\kappa$, $b = \nabla\rho$, and let $D \subset B(0, R)$ be measurable, $R \geq 1$. Put $u = U_D$ and suppose*

$$u \geq -\delta \quad \text{on } B(0, 2R), \qquad |u| \leq \delta \quad \text{on } B(0, 2R) \setminus D.$$

Then $w = (u - \delta)_+$ vanishes off D, belongs to $H^1(\mathbb{R}^d)$, and, with $A = \{u > \delta\}$,

$$\int_{\mathbb{R}^d} |\nabla w + a\mathbf{1}_A b|^2 + a^2 \int_{D\setminus A} |b|^2 \leq C\delta R^{d-1}. \tag{5.1}$$

In particular $|D \setminus A| \leq C\delta R^{d-1}$.

Proof. The potential is continuous, tends to zero at infinity, and is harmonic off $\overline{D}$. The maximum principle extends $|u| \leq \delta$ from the annulus to the exterior of $B(0, R)$. The potential equation is $\operatorname{div}(\nabla u + a\mathbf{1}_D b) = 0$. Since $\mathbf{1}_D b \in L^2$, the usual Fourier multiplier estimate gives $\nabla u \in L^2$; hence the compactly supported truncation w is in H^1. Testing the equation with w, using $A \subset D$, gives

$$\int |\nabla w|^2 = -a \int b \cdot \nabla w. \tag{5.2}$$

Choose a smooth χ equal to one on $B(0, R)$, zero off $B(0, 2R)$, with $|\nabla\chi| \leq C/R$ and $|\Delta\chi| \leq C/R^2$. Put $v = u - w$, so $|v| \leq \delta$. The equation tested with $\chi\rho$, together with (5.2), gives the exact identity

$$\int |\nabla w + a\mathbf{1}_D b|^2 = a \int b \cdot \nabla w + a^2 \int_D |b|^2 = -a \int \nabla v \cdot \nabla(\chi\rho) = a \int v\, d\Delta(\chi\rho).$$

The convexity measure $\mu = \Delta\rho$ has $\mu(B(0, 2R)) \leq CR^{d-1}$. Since $\Delta(\chi\rho) = \chi\mu + 2b \cdot \nabla\chi + \rho\Delta\chi$, its total variation is at most CR^{d-1}. Integration by parts is justified by uniform and local H^1 mollification of the continuous function v. Bounding the right side by $C\delta R^{d-1}$ and splitting the square over A and $D \setminus A$ proves the estimate. Finally $|b| \geq c$ almost everywhere, by Euler's identity and $\rho(x) \geq c|x|$. □

Corollary 5.2 (Exact rigidity)**.** *For bounded D, if $U_D \geq 0$ and $U_D = 0$ on D^c, then, up to null sets, $D = rK$ for some $r \geq 0$ and $U_D = 2d\kappa(r - \rho)_+$.*

Proof. If $|D| = 0$, take $r = 0$. Otherwise take $\delta = 0$. Then $D = \{u > 0\}$ up to null sets and $\nabla u = -a\mathbf{1}_{\{u>0\}}\nabla\rho$. Thus u is Lipschitz and $u + a\rho$ is constant on each connected component of its positivity set. On such a component write $u = a(r - \rho)$. The segment from any point of the component toward the origin cannot leave it: at a first exit its value would still be strictly positive. Every component therefore contains the origin, so there is only one. Its boundary lies in $\{\rho = r\}$; following each radial segment now shows that the component is rK. □

Corollary 5.3 (Few sites have small local time)**.** *On the same event, simultaneously for every* $t \geq 0$,

$$\#\{x \in A_n : \ell_n(x) \leq t\} \leq C(t + h)r^{d-1}, \qquad h = \begin{cases} \sqrt{r \log n}, & d = 2, \\ \log n, & d \geq 3. \end{cases}$$

Proof. The one-sided estimate gives $U_{D_n} \geq -Ch$ everywhere and $|U_{D_n}| \leq Ch$ off D_n. At a site with $\ell_n(x) \leq t$, (3.2) and Young's inequality give $U_{D_n}(x) \leq C(t+h)$; in the plane use $(\log n)^2 \leq Ch$. The same bound holds on its cell by the cell modulus. Apply Lemma 5.1 with $R = Cr$ and truncation threshold $C(t + h)$. All the indicated cells belong to its discarded set $D_n \setminus \{U_{D_n} > C(t+h)\}$, whose volume is at most $C(t + h)r^{d-1}$. This proves the assertion without assuming a uniform error of order h at highly occupied sites. □

5.1. **A sharp Sobolev cone inequality.** The calibration controls the defect $\nabla w + a\nabla\rho$ on the positivity set of w. Nguyen's integral inequality bounds the mass of w in terms of this defect and the volume of that set. The following form allows every convex gauge and retains the directional information in the defect. For the ℓ^1 ball, $a = 2\theta$ and $R_K = 1$, even its Euclidean-energy version improves the error coefficient by the factor $\sqrt{d/(d+2)}$.

Proposition 5.4 (A sharp cone inequality for every gauge)**.** *Let K be as above, $V = |K|$, and let $h_K(z) = \sup_{x\in K} x \cdot z$ be its support function. If $w \in H^1(\mathbb{R}^d)$ is nonnegative and $M = |\Omega| < \infty$, where $\Omega = \{w > 0\}$, then for every $a > 0$,*

$$\int w \leq \frac{aV^{-1/d}}{d+1} M^{1+1/d} + \frac{V^{-1/d}}{\sqrt{d(d+2)}} M^{1/2+1/d} \big\| h_K(-\nabla w - a\nabla\rho) \big\|_{L^2(\Omega)}. \tag{5.3}$$

The coefficients in (5.3) are sharp: equality holds for every $r > 0$ and $c \geq 0$ when $w = a(r - \rho)_+ + \frac{c}{2}(r^2 - \rho^2)_+$. Writing $R_K = \max_{x\in K} |x|$, the last norm can be replaced by $R_K\|\nabla w + a\nabla\rho\|_{L^2(\Omega)}$.

Proof. Assume $M > 0$ and put $\mu(t) = |\{w > t\}|$. For $0 < s < M$, the rank bound $|\{x \in \Omega : \mu(w(x)) > s\}| \leq M - s$ gives, by layer cake,

$$\int_\Omega \mu(w)^q \leq \frac{M^{q+1}}{q+1} \qquad (q > 0). \tag{5.4}$$

Indeed, the set in the rank bound is the increasing union of $\{0 < w \leq t\}$ over levels with $\mu(t) > s$. The anisotropic isoperimetric inequality and weighted coarea formula [17, (1.4), (2.22)] give

$$dV^{1/d} \int w \leq \int_0^\infty \mu(t)^{1/d} P_K(\{w > t\})\, dt = \int_\Omega \mu(w)^{1/d} h_K(-\nabla w).$$

Here P_K uses the outward normal, accounting for the minus sign; the argument therefore does not require $K = -K$. The Sobolev and coarea statements apply since $w, \nabla w \in L^1$, by Cauchy–Schwarz on Ω and $\nabla w = 0$ almost everywhere on its complement. Since $h_K(\nabla\rho) = 1$ almost everywhere, subadditivity gives

$$h_K(-\nabla w) \leq a + h_K(-\nabla w - a\nabla\rho).$$

Use (5.4) with $q = 1/d$ for the first term and, after Cauchy–Schwarz, with $q = 2/d$ for the second. Division by $dV^{1/d}$ proves (5.3). For the displayed equality family the defect is $c\rho$ on rK; the identities $\int_{rK} \rho^j = dVr^{d+j}/(d + j)$ verify equality directly. Finally $h_K(z) \leq R_K|z|$ proves the Euclidean-energy form. □

6. Limit laws

Lemma 6.1 (One covariance identity)**.** *For a lattice function f, let*

$$M_n^f = f(X_n) - f(0) - \sum_{j<n} \mathbb{E}[f(X_{j+1}) - f(X_j) \mid \mathcal{F}_j].$$

Write

$$\Gamma(f,h)(x) = \frac{1}{2d} \sum_{|e|=1} [f(x+e) - f(x)][h(x+e) - h(x)] - \Delta f(x)\Delta h(x),$$
$$o_f(x) = \max_{|e|=|e'|=1} |f(x+e) - f(x+e')|.$$

Then

$$\langle M^f, M^h \rangle_n = \sum_x \ell_n(x)\Gamma(f,h)(x) + R_n(f,h), \qquad |R_n(f,h)| \le C \sum_{x\in A_n} o_f(x)o_h(x). \tag{6.1}$$

Proof. Repeated departures contribute exactly $\Gamma(f,h)(x)$. At a first departure, covariance under the conditional cookie law differs from covariance under the uniform law by at most $Co_f(x)o_h(x)$: subtract one neighboring value from each variable, and bound the two product expectations and the two products of means. Each site contributes at most once. The argument is conditional on the past, so adaptive cookie laws require no change. □

Proposition 6.2 (Joint polynomial fluctuations)**.** *Set $b(x) = \kappa\lambda(x)\xi(x)$ and*

$$Z_n = X_n + \sum_{x\in A_n} b(x), \qquad Q_n = |X_n|^2 - n + 2\kappa S_n.$$

These are martingales. With $c_K = 2d\kappa V/(d+1)$ and $m_K = \int_K z\,dz$, their joint limit is

$$\left(\frac{Z_n}{\sqrt{n}}, \frac{Q_n}{r_n^{(d+3)/2}}\right) \Longrightarrow N\left(0, \begin{pmatrix} I_d/d & \dfrac{4\kappa m_K}{(d+2)\sqrt{c_K}} \\ \dfrac{4\kappa m_K^{\mathsf{T}}}{(d+2)\sqrt{c_K}} & \dfrac{8\kappa J_K}{d+3} \end{pmatrix}\right). \tag{6.2}$$

For either sign, almost surely,

$$\limsup_n \frac{\pm Q_n}{r_n^{(d+3)/2}\sqrt{2\log\log n}} = \sqrt{\frac{8\kappa J_K}{d+3}}, \qquad \limsup_n \frac{\pm v\cdot Z_n}{\sqrt{2n\log\log n}} = \frac{|v|}{\sqrt{d}} \quad (v\in\mathbb{R}^d).$$

For symmetric K the two components of the Gaussian limit are independent.

Proof. Euler's identity gives $x\cdot b(x) = \kappa\lambda(x)\rho(x)$; thus $Z = M^x$ and $Q = M^{|x|^2}$. Since

$$\Gamma(x_i,x_j) = \delta_{ij}/d, \qquad \Gamma(x_i,|x|^2) = 2x_i/d, \qquad \Gamma(|x|^2,|x|^2) = 4|x|^2/d,$$

Lemma 6.1 gives

$$\langle Z\rangle_n = \frac{n}{d} I_d + O(|A_n|), \tag{6.3}$$

$$\langle Z, Q\rangle_n = \frac{2}{d}\sum_x \ell_n(x)x + O\left(\sum_{x\in A_n} |x|\right), \tag{6.4}$$

$$\langle Q\rangle_n = \frac{4}{d}\sum_x \ell_n(x)|x|^2 + O\left(\sum_{x\in A_n} |x|^2\right). \tag{6.5}$$

The neighbor oscillations are $O(1)$ and $O(|x|)$, respectively; in particular the quadratic correction vanishes at $x = 0$. The qualitative shape theorem and cone profile turn these sums into Riemann integrals. Homogeneity gives

$$\int_K (1-\rho(z))z\,dz = \frac{m_K}{d+2}, \qquad \int_K (1-\rho(z))|z|^2\,dz = \frac{J_K}{d+3}.$$

The covariance errors are smaller by a factor r_n^{-1}. Dividing the three brackets by n, $\sqrt{n}\,r_n^{(d+3)/2}$ and r_n^{d+3} proves their almost-sure limits in (6.2). The increment bounds $|\Delta Z_j| \le 2$ and $|\Delta Q_j| \le 4|X_j|$, together with $\max_{j\le n}|X_j| = O(r_n)$ eventually almost surely, imply Lindeberg after stopping at a deterministic multiple of r_n. The stopping has probability tending to zero by the coarse estimate. The multivariate martingale CLT gives (6.2) [21, Corollary 3.1]. The same increment bounds verify Stout's condition, because their ratios to the respective $\sqrt{\langle M\rangle_n/\log\log\langle M\rangle_n}$ are $O(\sqrt{\log\log n/n})$. This gives both LILs [43]. If $K = -K$, then $m_K = 0$, proving the independence assertion. □

Proof of Theorem 1.1(iv). In the identity $2\kappa(S_n - n/(2\kappa)) = Q_n - |X_n|^2$, the endpoint is $O(r_n^2)$ and hence negligible on the CLT and LIL scales for $d \ge 2$. Proposition 6.2 gives both asserted laws for S_n. They also give $S_n \sim n/(2\kappa)$ almost surely. The derivative of $((d+1)s/(dV))^{1/(d+1)}$ at $s = n/(2\kappa)$ is $1/(dVr_n^d)$, which gives the stated laws for $R_{\mathrm{mom}}(n)$. □

Theorem 6.3 (Local times centered by the cookie potential)**.** *Under the assumptions of Theorem 1.1, use the Green kernel g of Lemma 2.2. Define the range-dependent potential and normalization by*

$$P_n(y) = \sum_{x\in A_n}\sum_{|e|=1}\left(\frac{1}{2d} - q_x(e)\right)g(x+e-y), \qquad s_n = \begin{cases}\sqrt{r_n\log r_n}, & d=2,\\ \sqrt{r_n}, & d\ge 3.\end{cases}$$

For every finite list $y_1,\dots,y_k$, $((\ell_n(y_i) - P_n(y_i))/s_n)_i$ tends to a centered Gaussian vector of covariance

$$\Sigma_{ij} = \begin{cases} 16\kappa/\pi, & d=2,\\ 2d\kappa(2G(y_i-y_j) - \mathbf{1}_{\{y_i=y_j\}}), & d\ge 3.\end{cases}$$

Write $\Sigma(y,y)$ for the corresponding diagonal variance. At each fixed site, for either sign, almost surely,

$$\limsup_n \frac{\pm(\ell_n(y) - P_n(y))}{s_n\sqrt{2\log\log n}} = \sqrt{\Sigma(y,y)}.$$

Both assertions also hold with $U_{D_n}(y)$ in place of $P_n(y)$. A deterministic cone requires the centering estimate below.

Proof. The Dynkin martingale M^y for $g_y(x) = g(x-y)$ satisfies

$$\ell_n(y) - P_n(y) = -M_n^y + g(X_n - y) - g(-y).$$

The endpoint is $O(\log n)$ in $d = 2$ and $O(1)$ in $d \ge 3$. Apply Lemma 6.1 with $f = g_y$ and $h = g_z$. The bound $|g(x+e) - g(x)| \le C(1+|x|)^{1-d}$ makes the first-cookie correction $O_{y,z}(\log r_n)$ in the plane and $O_{y,z}(1)$ in higher dimensions. Thus

$$\langle M^y, M^z\rangle_n = \sum_x \ell_n(x)\Gamma(g_y,g_z)(x) + O_{y,z}(1+\log r_n).$$

For $d \ge 3$, summability, the cone profile and summation by parts give $r_n^{-1}\langle M^y, M^z\rangle_n \to 2d\kappa(2G(y-z) - \mathbf{1}_{\{y=z\}})$. In $d = 2$, the covariance kernel is $2/(\pi^2|x|^2) + O_{y,z}(|x|^{-3})$, whose sum over a ball of radius r is $(4/\pi)\log r + O(1)$. The constant $4\kappa r_n$ of the cone gives $(16\kappa/\pi)r_n\log r_n$; the term involving $\rho(x)$ and the change from a Euclidean ball to a gauge ball cost only $O(r_n)$. The profile error costs $o(r_n\log r_n)$. These are the bracket limits. Bounded increments allow the multivariate martingale CLT and the scalar LIL. The potential representation gives $P_n(y) = U_{D_n}(y) + O(\log n)$ eventually almost surely, which proves the assertion about continuum centering. □

Lemma 6.4 (Centering at a fixed site)**.** *Assume the Euclidean inner-radius and volume estimates* (1.12)*, and put*

$$q_n = \begin{cases} \sqrt{\log n / r_n}, & d = 2, \\ \log n / r_n, & d \geq 3, \end{cases} \qquad w_n = \max\{|R_{\text{in}}(n) - r_n|, |R_{\text{out}}(n) - r_n|\} + 2\sqrt{d}.$$

For every fixed y, almost surely for all sufficiently large n,

$$U_{D_n}(y) - 2d\kappa(r_n - |y|) - \frac{Q_n}{\omega_d r_n^d} = O_y(1 + q_n w_n). \tag{6.6}$$

Consequently the qualitative outer shape $w_n = o(r_n)$ already suffices for every deterministic fixed-site limit law.

Proof. Write $\nu = \mathbf{1}_{D_n} - \mathbf{1}_{B(0,r_n)}$. It is supported on $\{v : ||v| - r_n| \leq w_n\}$, and the inner-volume estimate gives $\|\nu\|_1 \leq C r_n^d q_n$. On this annulus, eventually,

$$\left| |v|^{1-d} - r_n^{-d}|v| \right| \leq C r_n^{-d} ||v| - r_n|.$$

The radial kernel at zero is $(2\kappa/\omega_d)|v|^{1-d}$. Integration against ν therefore gives

$$U_{D_n}(0) - 2d\kappa r_n = \frac{2\kappa}{\omega_d r_n^d} \left(\int_{D_n} |v| \, dv - \frac{n}{2\kappa} \right) + O(q_n w_n).$$

Cell replacement costs $O(r_n^d)$, as does the weighted correction:

$$\left| S_n - \sum_{x \in A_n} |x| \right| \leq C r_n \sum_{|x| \leq C r_n} |\lambda(x) - 1| = O(r_n^d).$$

The parenthesis is therefore $(Q_n - |X_n|^2)/(2\kappa) + O(r_n^d)$. Since $|X_n|^2/r_n^d = O(1)$, this proves the claim at zero. For fixed y, the difference between the kernels at y and zero is $O_y(r_n^{-d})$ on the support of ν; its integral is $O_y(1)$. The ball potential changes by exactly $-2d\kappa|y|$, proving (6.6). □

Deterministic centering in Theorem 1.2(ii). The qualitative shape gives $w_n = o(r_n)$, hence $q_n w_n = o(\sqrt{r_n \log n})$ in the plane and $q_n w_n = o(\log n) = o(\sqrt{r_n})$ in higher dimensions. The moment LIL gives, almost surely,

$$Q_n / r_n^d = O\left(r_n^{(3-d)/2} \sqrt{\log \log n} \right).$$

Thus both terms in (6.6) are negligible on the fixed-site CLT scale, and therefore also on its LIL scale. Theorem 6.3 proves the claimed joint Gaussian and signed LIL laws. No rate for the outer shape is needed here. □

7. Sharpness and fluctuation lower bounds

The lower bounds use a coarse filled shape and the covariance identity, not the sharp Euclidean radius estimates. We first record the coarse input for every convex gauge. Write $r = r_n$ and $L = \log n$. For every fixed $p > 0$, there is an event Ω_n with $\mathbb{P}(\Omega_n^c) \leq C_p n^{-p}$ on which

$$crK \subset D_n \subset B(0, C_0 r), \qquad \max_x \ell_n(x) \leq Cr, \qquad \|\widetilde{\ell}_n - U_{D_n}\|_\infty \leq \delta, \quad \delta = C\sqrt{r} \log n. \tag{7.1}$$

The filled inclusion and local-time bound are part of Proposition 2.5; the Green-martingale estimate gives the uniform error. Below take $p = 10$.

Lemma 7.1 (One common bulk height)**.** *On Ω_n, let $a = 2d\kappa$ and $v_n = U_{D_n}(0)$. Then*

$$cr \leq v_n \leq Cr, \qquad |\ell_n(x) - v_n| \leq C \min\{|x|, r\} + C\delta \quad (x \in \mathbb{Z}^d). \tag{7.2}$$

For all y, z in a sufficiently small fixed multiple of $B(0, r)$,

$$|U_{D_n}(z) - U_{D_n}(y) + a(\rho(z) - \rho(y))| \leq C|z - y|. \tag{7.3}$$

Also $\ell_n(x) \geq cr$ on a sufficiently small fixed multiple of $B(0,r)$ for large n.

Proof. Write $D_n = bK \cup E$ with $b \geq cr$. Inside bK, convexity makes the integrand of U_E nonnegative, while $U_{bK} = a(b-\rho)_+$. This gives the lower bounds for v_n and, using $\delta = o(r)$, for ℓ_n on a smaller ball. The potential-volume bound gives $v_n \leq Cr$. The excess set is a distance of order r from this smaller ball and has volume $O(r^d)$; the differentiated potential kernel is $O(r^{-d})$ there. Hence $|\nabla U_E| \leq C$, proving (7.3) and $|U_{D_n}(x) - v_n| \leq C|x|$ near zero. Outside the smaller ball, $\min\{|x|, r\} \geq cr$, and both ℓ_n and v_n are at most Cr. This proves (7.2) globally. □

7.1. **One exponential-martingale lemma.**

Lemma 7.2. *Fix $0 < c_0 \leq C_0$. Let $S^1, \dots, S^m$ be real martingales starting at zero with increments bounded by b. Suppose an event Ω, of probability at least $1-\alpha$, satisfies $c_0 \leq \langle S^i \rangle_n \leq C_0$ for every i. There are $c, C > 0$, depending only on c_0, C_0, such that if $\beta \geq C$ and $\beta b \leq c$, then*

$$\mathbb{P}(\pm S^i_n \geq c\beta) \geq \tfrac{1}{4} e^{-C\beta^2} - \alpha.$$

If additionally $|\langle S^i, S^j \rangle_n| \leq \delta$ on Ω for $i \neq j$ and $\beta^2\delta + \beta^3 b \leq 1$, then

$$\mathbb{P}\{\max_i S^i_n < c\beta\} \leq C\left(m^{-1}e^{C\beta^2} + \beta^2\delta + \beta^3 b + e^{C\beta^2}\sqrt{\alpha}\right).$$

Proof. Stop all martingales just before any bracket would exceed $2C_0$; this changes none of them on Ω. For a martingale S whose increments are at most b_1 and whose bracket is bounded by $8C_0$, put

$$K_S(s) = \sum_{j<n} \log \mathbb{E}(e^{s\Delta S_j} \mid \mathcal{F}_j), \qquad Z_S(s) = e^{sS_n - K_S(s)}.$$

Conditional Taylor expansion, for $|s|b_1 \leq 1/2$, gives

$$K_S(s) = \tfrac{1}{2}s^2\langle S \rangle_n + O(|s|^3 b_1), \quad \mathbb{E}Z_S(s) = 1, \quad \mathbb{E}Z_S(s)^2 \leq e^{Cs^2}.$$

For the last bound use $Z_S(s)^2 = Z_S(2s)e^{K_S(2s)-2K_S(s)}$. Apply these formulas to S^i and $S^i + S^j$, with $s = \beta$, and abbreviate $Z_i = Z_{S^i}(\beta)$. On Ω, $Z_i \geq 1/2$ implies $S^i_n \geq c\beta$, since $K_{S^i}(\beta) \geq c_0\beta^2/4$. Paley–Zygmund therefore gives the first assertion; use $-S^i$ for the opposite sign. For $i \neq j$,

$$Z_i Z_j = Z_{S^i+S^j}(\beta)\exp\{\beta^2\langle S^i, S^j \rangle_n + O(\beta^3 b)\}.$$

On Ω the exponential is at most $1 + C(\beta^2\delta + \beta^3 b)$; on its complement apply Cauchy–Schwarz and the second-moment bound. Consequently

$$\mathbb{E}Z_i Z_j \leq 1 + C(\beta^2\delta + \beta^3 b) + e^{C\beta^2}\sqrt{\alpha}.$$

The average $\overline{Z} = m^{-1}\sum_i Z_i$ has mean one and variance bounded by the right side of the asserted probability estimate. If $\overline{Z} \geq 1/2$ on Ω, some $S^i_n \geq c\beta$. Chebyshev, with the additional error α, finishes the proof. □

7.2. **Planar radii and width.**

In this subsection only, let $d = 2$ and $K = B(0,1)$; put $R_- = R_{\text{in}}(n)$, $R_+ = R_{\text{out}}(n)$, and $u_x = x/|x|$ for $x \neq 0$, with $u_0 = 0$. The polynomial martingales from Proposition 6.2 are

$$Q_n = |X_n|^2 - n + 2\kappa \sum_{x \in A_n} \lambda(x)|x|, \qquad Z_n = X_n + \kappa \sum_{x \in A_n} \lambda(x) u_x.$$

On Ω_n the sparse-weight condition gives

$$\left| S_n - \sum_{x \in A_n} |x| \right| \leq Cr^2, \qquad \left| \sum_{x \in A_n} (\lambda(x) - 1)u_x \right| \leq Cr. \tag{7.4}$$

The coarse filled ball and positivity of the excess potential imply $\ell_n(x) \geq cr$ on a smaller ball. Equations (6.3)–(6.5) consequently give, on Ω_n,

$$cr^5 \leq \langle Q \rangle_n \leq Cr^5, \qquad \langle v \cdot Z \rangle_n = |v|^2 n/2 + O(|v|^2 r^2). \tag{7.5}$$

The first-cookie errors are $O(r^4)$ and $O(r^2)$, respectively. These coarse estimates suffice for every polynomial lower tail below; the exact almost-sure bracket limits and LIL constants were already proved in Proposition 6.2.

The unweighted moment radius

$$T_n = \left(\frac{3}{2\pi} \sum_{x \in A_n} |x| \right)^{1/3}$$

satisfies $R_- \leq T_n + C$ and $R_+ \geq T_n - C$ by comparison with unit cells. The weighted moment radius differs from T_n by $O(1)$ on Ω_n and eventually almost surely, by (7.4) and the shape theorem. The moment LIL in Theorem 1.1 therefore proves

$$\limsup_n \frac{r - R_-}{\sqrt{r \log\log n}}, \quad \limsup_n \frac{R_+ - r}{\sqrt{r \log\log n}} \geq \frac{1}{\sqrt{10\pi\kappa}} \quad \text{almost surely.} \tag{7.6}$$

To obtain polynomial lower tails, stop Q on exiting $B_{C_0 r}$ and apply Lemma 7.2 to $Q/r^{5/2}$. Its increments are bounded by $4C_0 r^{-3/2}$, and its bracket lies between positive constants on Ω_n by (7.5). Choose $\beta = a_p \sqrt{\log n}$, with $a_p > 0$ small enough that $Ca_p^2 < \min(p, 10)/2$. Each sign then has probability exceeding n^{-p} after intersecting with Ω_n and subtracting its failure probability once more, for large n. Since $|X_n|^2$ and the defect in (7.4) are $O(r^2) = o(r^{5/2}\sqrt{\log n})$, the identity for Q and factoring $T_n^3 - r^3$ yield, with a suitable $c_p > 0$,

$$\mathbb{P}\{r - R_- \geq c_p \sqrt{r \log n}\} \geq n^{-p}, \qquad \mathbb{P}\{R_+ - r \geq c_p \sqrt{r \log n}\} \geq n^{-p}. \tag{7.7}$$

For the width set $V_n = \sum_{x \in A_n} u_x$. The sum over the lattice disk $\{|x| < R_-\}$ vanishes by symmetry. When $R_-, R_+ \asymp r$, integration of the positive part of any directional component over the remaining annulus gives

$$|V_n| \leq R_+^2 - R_-^2 + Cr, \qquad |Z_n| \leq \kappa(R_+^2 - R_-^2) + Cr. \tag{7.8}$$

Indeed, $\int_0^{2\pi} (\cos\theta)_+ \, d\theta = 2$; replacing each unit vector by its values in the corresponding cell costs $O(r)$, as does the weighted correction. The martingale $e_1 \cdot Z$ has increments at most two. Apply Lemma 7.2 to $(e_1 \cdot Z)/\sqrt{n}$, using its bracket bounds on Ω_n from (7.5). Equation (7.8) and $n = (4\pi\kappa/3)r^3$ give

$$\mathbb{P}\{R_+ - R_- \geq c_p \sqrt{r \log n}\} \geq n^{-p}. \tag{7.9}$$

The same bracket limit, the bounded-increment martingale LIL, and $R_- + R_+ \sim 2r$ give

$$\limsup_n \frac{R_+ - R_-}{\sqrt{r \log\log n}} \geq \sqrt{\frac{\pi}{3\kappa}} \quad \text{almost surely.} \tag{7.10}$$

Finally the vector martingale CLT yields $Z_n/\sqrt{n} \Longrightarrow N(0, I_2/2)$. On $\{R_+ - R_- \leq a\sqrt{r}\}$, (7.8) gives $|Z_n|/\sqrt{n} \leq \sqrt{3\kappa/\pi}\, a + o(1)$ almost surely. Since $\mathbb{P}\{|N(0, I_2/2)| \leq t\} = 1 - e^{-t^2}$,

$$\limsup_n \mathbb{P}\{R_+ - R_- \leq a\sqrt{r}\} \leq 1 - e^{-3\kappa a^2/\pi} \qquad (a \geq 0). \tag{7.11}$$

Corollary 7.3 (A joint planar small-ball bound)**.** *In dimension two with $K = B(0,1)$, let $\sigma^2 = 1/(20\pi\kappa)$ and let Φ be the standard Gaussian distribution function. For every $\alpha, \beta \geq 0$,*

$$\begin{aligned} \limsup_n \mathbb{P}\{R_{\mathrm{in}}(n) \geq r_n - \alpha\sqrt{r_n}, \ R_{\mathrm{out}}(n) \leq r_n + \beta\sqrt{r_n}\} \\ \leq [\Phi(\beta/\sigma) - \Phi(-\alpha/\sigma)][1 - e^{-3\kappa(\alpha+\beta)^2/\pi}]. \end{aligned} \tag{7.12}$$

Proof. Proposition 6.2 and the delta method give independent Gaussian limits for $(R_{\mathrm{mom}}(n) - r_n)/\sqrt{r_n}$ and $Z_n/\sqrt{n}$, of laws $N(0,\sigma^2)$ and $N(0, I_2/2)$. The cell comparison above and the $O(1)$ weighted correction show that the event in the assertion forces the first quantity into $[-\alpha,\beta]+o(1)$. Equation (7.8) forces $|Z_n|/\sqrt{n} \le \sqrt{3\kappa/\pi}(\alpha+\beta)+o(1)$. The Gaussian boundaries have probability zero. The joint CLT gives the product bound. □

7.3. **The maximum of the bulk local times.**

Theorem 7.4 (Bulk oscillations for every convex gauge)**.** *Under the general model assumptions, there is $c>0$ such that for all sufficiently large n, with probability at least $1-n^{-c}$,*

$$\operatorname{osc}_{|x|\le\sqrt{r_n}}\big(\ell_n(x)+2d\kappa\rho(x)\big) \ge c\begin{cases}\sqrt{r_n}\log n, & d=2,\\ \sqrt{r_n\log n}, & d\ge 3.\end{cases} \tag{7.13}$$

Here the oscillation is the maximum minus the minimum over lattice sites. In particular the same orders, with a different constant, lower-bound

$$\max_{|x|\le\sqrt{r_n}} |\ell_n(x) - 2d\kappa(r_n-\rho(x))| \tag{7.14}$$

with the same probability. This proves Theorem 1.1(v) *and includes the Euclidean bulk maximum assertion.*

Proof. Put $a=2d\kappa$ and work on the coarse event (7.1) with failure probability $O(n^{-10})$. Put

$$k=m=\lfloor r^{1/8}\rfloor, \qquad y_i = i\lceil r^{1/4}\rceil e_1 \ (1\le i\le m), \qquad h=ke_1, \qquad f_i = g_{y_i+h}-g_{y_i}.$$

The same dipoles are used in every dimension. All poles lie in $B(0,\sqrt{r})$, and in fact in $B(0,Cr^{3/8})$. Set $\alpha_r=\log r$ in dimension two and $\alpha_r=1$ in higher dimensions, $s_r^2 = r\alpha_r$, and $S^i = -M^{f_i}/s_r$.

We claim that on the coarse event

$$c_0 \le \langle S^i\rangle_n \le C_0, \qquad |\langle S^i, S^j\rangle_n| \le Cr^{-1/4} \quad (i\ne j). \tag{7.15}$$

For clarity, the leading height in this calculation is the random $v_n = U_{D_n}(0)$; a deterministic cone approximation is unnecessary. Let $\mathcal{E}_{ij} = \sum_x \Gamma(f_i,f_j)(x)$. Green-kernel estimates and $2ab\le a^2+b^2$ give

$$\sum_x |\Gamma(f_i,f_j)(x)| \le C\alpha_r,$$

$$\sum_x \min\{|x|,r\}|\Gamma(f_i,f_j)(x)| \le C\sqrt{r}\,\alpha_r.$$

Here are the details behind these summations. In dimensions at least three it suffices to dominate each dipole by its two monopoles, whose squared increments are bounded by $C(1+|x-y|)^{2-2d}$. Their total mass is bounded and their first truncated moment is $O(\log r)$ in dimension three and $O(1)$ above it. In the plane, within distance $3k$ of a pole, use the sum of its two reciprocal-distance bounds. Outside this region, cancellation gives an increment bound $Ck(1+|x-y_i|)^{-2}$. The squared sum is $O(\log r)$, and its first moment about y_i is $O(k\log r)$. Translating by $|y_i|\le Cr^{3/8}$ proves the assertions. The finitely supported Laplacian terms obey the same bounds.

By (6.1) and (7.2),

$$\left|\langle S^i,S^j\rangle_n - \frac{v_n\mathcal{E}_{ij}}{r\alpha_r}\right| \le C\left(\frac{\delta}{r}+r^{-1/2}+r^{-1}\right) \le Cr^{-1/2}\log n = o(r^{-1/4}).$$

The first-cookie term is $O(\alpha_r)$ before normalization, by the same increment-square estimates. Summation by parts now gives an exact formula: if $z=y_i-y_j$, then

$$\mathcal{E}_{ij} = 2\{g(z+h)+g(z-h)-2g(z)\} - 2\mathbf{1}_{\{i=j\}}. \tag{7.16}$$

For the diagonal this equals $4(g(h)-g(0))-2$. In the plane it is asymptotic to $(8/\pi)\log k$, hence to $(1/\pi)\log r$; in higher dimensions it tends to $4G(0)-2>0$. For $i\neq j$, the supports of the two Laplacians are disjoint, and the Green asymptotics in the second difference give

$$|\mathcal{E}_{ij}| \le Ck^2|y_i-y_j|^{-d} \le Cr^{-(d-1)/4}.$$

Together with $cr \le v_n \le Cr$, this proves (7.15).

The increments of S^i are bounded by $Cr^{-1/2}$. Apply Lemma 7.2 with $\beta=\varepsilon\sqrt{\log r}$, taking $\varepsilon>0$ sufficiently small. Since $m \asymp r^{1/8}$ and the cross brackets are $O(r^{-1/4})$, its failure bound is at most $Cr^{-1/16}$. Thus for some i, except with this probability, $-M_n^{f_i} \ge cs_r\sqrt{\log r}$.

Finally the Green representation and (7.3) give

$$\ell_n(y_i+h)-\ell_n(y_i)+a\big(\rho(y_i+h)-\rho(y_i)\big) = -M_n^{f_i}+O(k+\log n).$$

The error is $o(s_r\sqrt{\log r})$. The left side is the difference of two values whose oscillation appears in (7.13). Absorb the polynomial failure probability and use $r \asymp n^{1/(d+1)}$ to finish the proof. □

Remark 7.5. The window $\sqrt{r_n}$ can be replaced by r_n^η for any fixed $\eta>0$, with constants depending on η. For $0<\eta\le 1/2$, take $k=m=\lfloor r^{\eta/4}\rfloor$ and pole spacing $\lceil r^{\eta/2}\rceil$. The poles are then within $Cr^{3\eta/4}<r^\eta$, the planar diagonal energy is $\asymp \eta\log r$, and the off-diagonal energies are $O(r^{-(d-1)\eta/2})$. The common-height error is still $O(r^{-1/2}\log n)$. A sufficiently small multiple of $\sqrt{\log r}$ in Lemma 7.2 gives the same lower orders and a polynomial failure probability. For $\eta>1/2$ use inclusion of windows.

7.4. **Logarithmic radii under ellipticity.** Return to $K=B(0,1)$, with $R_-=R_{\mathrm{in}}(n)$ and $R_+=R_{\mathrm{out}}(n)$. For this paragraph assume additionally $q_x(e)\ge p_0>0$ for every x,e; later departures already have probability $1/(2d)$. There are $c,C,h_0>0$ such that, for $h_0\le h\le r/8$,

$$\mathbb{P}\{R_- \ge r-h,\ R_+ \le r+h\} \le e^{-cr^{d-1}e^{-Ch}}, \tag{7.17}$$

$$\mathbb{P}\{R_+ \le r+h\} \le e^{-cr^{d-1}e^{-Ch}} + e^{-cr^{d-3}h^2}. \tag{7.18}$$

To prove these estimates, from any x with $|x|>r-2h$ take $T=\lceil 4h\sqrt{d}\rceil$ steps in the signed coordinate of largest $|x_i|$. The Euclidean distance increases at speed at least $1/\sqrt{d}$, so this path exits B_{r+h}; its conditional probability is at least p_0^T after decreasing p_0 if necessary. Among k distinct arrivals at sites of any fixed $\Lambda\subset\{|x|>r-2h\}$ by time n, at least $\lfloor (k-T)/(T+1)\rfloor$ disjoint such trials can be started and completed by time n. Successive conditioning bounds their joint failure probability by $(1-p_0^T)^{\lfloor (k-T)/(T+1)\rfloor}$. If $R_-\ge r-h$, the annulus $r-2h<|x|<r-h-1$ supplies $k\ge cr^{d-1}h$ sites, proving (7.17).

For (7.18), let $\Lambda=\{r-2h<|x|\le r+h\}$. The same trials apply if $|A_n\cap\Lambda|\ge c_1r^{d-1}h$. Otherwise, on $R_+\le r+h$, cell comparison and the defect bound give

$$\sum_{x\in A_n}\lambda_x|x| \le \frac{n}{2\kappa}-c_2r^dh+2r|A_n\cap\Lambda|+Cr^d.$$

Choose c_1 small and then h_0 large. The identity $Q_n=|X_n|^2-n+2\kappa\sum\lambda_x|x|$ forces $Q_n\le -cr^dh$; the endpoint $|X_n|^2$ is absorbed by h_0 in $d=2$ and by large r in higher dimensions. Stop Q on leaving B_{r+h}. Its increments are bounded by $4(r+h)$, so Azuma's inequality gives $\mathbb{P}\{Q_n\le -cr^dh,\ R_+\le r+h\}\le e^{-cr^{d-3}h^2}$, proving the second estimate. Taking $h=c\log n$ with $c>0$ sufficiently small and applying Borel–Cantelli gives, almost surely for all sufficiently large n,

$$\max(r-R_-,R_+-r)>c\log n; \qquad R_+-r>c\log n \quad \text{if } d\ge 3.$$

The ellipticity assumption is used only in the prescribed-path trials and is not needed for any preceding lower bound.

Draft and verification scope. This is a new mathematical draft. The Lean certificate accompanying [9] is pinned to that paper and does not certify the generalized statements here.

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